WORKFORCE PLANNER

Staffing the Presidents' Day week

A forecast-driven labor plan for three warehouses, and the tool that runs it

Costco workforce planning case
Texas MISA Fall 2026 case competition

Team name to be added before submission
Planning week: Thursday, Feb 12 to Wednesday, Feb 18, 2026
This version was built from the analysis on Thursday, Nov 5, 2026.

> **Simulated data**
> All data in this report is simulated for the Texas MISA case competition. It is not Costco data, and it does not represent Costco wages, staffing levels, schedules or service performance.

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Summary

This report plans labor for one hard week, Thursday, Feb 12 to Wednesday, Feb 18, 2026, at three warehouses and under three sets of conditions. Plans are judged on service first, then labor cost, then whether they can be staffed.

Our answer: staff every block (one warehouse, department, day and shift) to at least a 95% chance of meeting its service target. We forecast demand, learn from history how service responds to workload, and allow for callouts and forecast error. Every block in every scenario stays within the staff available. In the expected week (BASE):

Expected service	Labor cost	Blocks below the goal	Blocks over capacity
95.6%	**$377,504**	**25**	**0**
Draft: 68.5%	+14.7% against the draft	of 180 open; draft: 135	Draft: 12 in DISRUPT

Table 1. The plan against the manager's draft in each scenario. Expected service is the average chance of meeting target across open blocks, from our risk model.

Scenario	Draft service	Plan service	Draft cost	Plan cost	Change	Below 95%
BASE	68.5%	**95.6%**	$329,108	$377,504	+14.7%	25
PEAK	58.3%	**87.0%**	$329,108	$387,926	+17.9%	45
DISRUPT	66.6%	**89.5%**	$326,367	$366,649	+12.3%	39

Most of the gain comes from putting hours in the right places. Only moving hours, at about the draft's cost, lifts BASE expected service from 68.5% to 73.4%. The rest is bought with extra holiday-weekend hours and an allowance for callouts.

What a warehouse leader should do:

1. **Run the plan that matches the week:** BASE by default, PEAK if the promotion or storm is confirmed, DISRUPT if Warehouse 2's illness or Warehouse 4's crew shortage is reported.
2. **Limit holiday PTO on Feb 14-16.** Keeping up to 5 people per block from holiday time off cuts BASE blocks below the goal from 25 to 6 and saves $4,381, as regular staff replace overtime.
3. **Approve overtime early and escalate the gaps.** Even at full capacity, BASE is 43 people short of the goal, mostly on Feb 14; PEAK is 190 short and DISRUPT 200. Only more available people can close these gaps.

The forecast matches the case's benchmark on normal days and differs around the holidays, where it applies the effects measured on the Martin Luther King Jr. (MLK) weekend. **The tool**, the Workforce Planner, runs on one laptop with no internet: it shows the plan for any scenario and warehouse, explains every block, re-plans live and exports the submission file (chapter 8).

1 The case

A warehouse club runs on a simple promise: low prices, full shelves and short lines. Labor is one of the biggest costs it controls, and it is also what keeps that promise. This chapter sets out the decision the case asks us to make.

The business problem

Each week, managers decide how many people to schedule in each department and shift. Today they mostly copy recent weeks. That works when next week looks like last week. It fails when demand moves, as it does around holidays, promotions, storms and illness. The case asks for a way to forecast what a week will need, turn that into staffing, balance cost against service, and explain the result to the people who must act on it.

Three departments, and why each matters

Table 2. The three departments in the case and the service measure each is judged on.

Department	What the staff do	Service measure	Target	Shared staff
Front End	Run the checkout lanes and the exit	Checkout wait	5 minutes or less	Cross-trained with Merchandising
Merchandising	Stock the floor and refill shelves	Replenishment response	60 minutes or less	Cross-trained with Front End
Pharmacy	Fill prescriptions and counsel members	Prescription wait	20 minutes or less	None: licensed staff only

Front End is what members feel first: a short staff at checkout means long lines at the busiest hours. **Merchandising** keeps the shelves full; when it falls behind, items run out on the floor and restocking spills into busy hours. **Pharmacy** has the least room to adjust, because its staff are licensed and can't be borrowed from another department. Each department counts its demand in its own workload units, so we never compare units across departments.

Front End and Merchandising share a small pool of cross-trained staff in each shift. Someone covering checkout can't also be stocking shelves, so the two departments must be planned together.

The planning week

The week runs Thursday, Feb 12 to Wednesday, Feb 18, 2026. It holds two events that six weeks of history can't show directly: **Valentine's Day** on Saturday, Feb 14, and **Presidents' Day** on Monday, Feb 16. Friday to Monday forms the Presidents' Day window, when holiday shopping and holiday time off both peak (Figure 1).

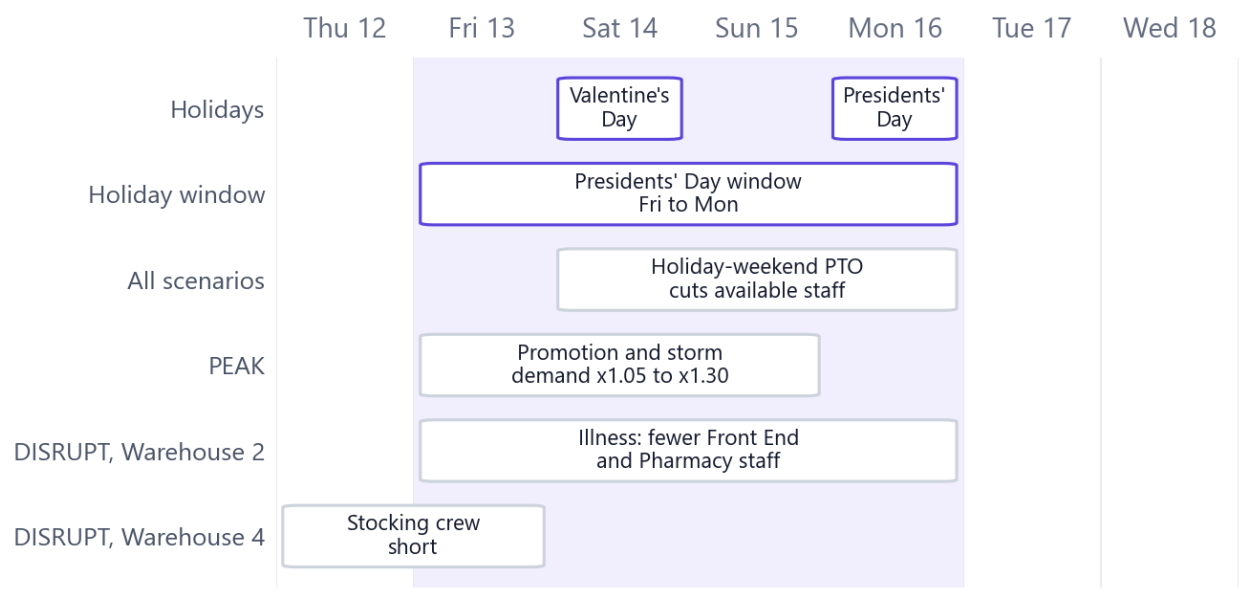


Figure 1. What happens on each day of the planning week, read from the case's planning grid. The shaded days are the Presidents' Day window. Time off over the holiday weekend applies in every scenario; PEAK and DISRUPT add their own changes.

We plan three warehouses: Warehouse 1 is medium-sized, and Warehouses 2 and 4 are large (there is no Warehouse 3). Each has the three departments and three shifts a day: opening, midday and evening. One warehouse, department, day and shift is a **block**, the unit everything in this report is planned in. That gives 189 blocks per scenario and 567 rows in the submission. Of those, 9 per scenario are closed, all of them Pharmacy shifts on the weekend.

The three scenarios

The case asks for a plan under each of three sets of conditions. All three are required.

Table 3. The three scenarios, as the planning grid applies them.

Scenario	What changes	When	Effect on staffing
BASE	Expected conditions for the week, including holiday-weekend time off	All week	Our forecast against the planned availability
PEAK	A member promotion launches and a winter-storm advisory brings stock-up trips. Demand rises by x1.05 to x1.30	Fri Feb 13 to Sun Feb 15	More demand, the same people
DISRUPT	An illness cluster at Warehouse 2 (about 30% of Front End out, one Pharmacy team member out per block, overtime halved), and a receiving crew shortage at Warehouse 4 (about 35% of the opening stocking crew)	Warehouse 2: Feb 13-16. Warehouse 4: Feb 12-13	The same demand, fewer people

What we hand in, and how it's judged

- **A presentation** of 12-15 minutes with a demo of the tool, covering the diagnosis, the forecast, the recommendation and all three scenarios.
- **The completed submission template:** the total employees for every block in every scenario, 567 rows, with 0 for closed blocks.
- **A working tool** a warehouse leader could use: it takes a scenario and shows staffing, labor cost and service risk.
- **Supporting analysis** showing the forecast method, the key assumptions (with ours marked) and cost and service against the draft. This report is that analysis.

Plans are judged on service first, then labor cost, then feasibility. Feasible means each block stays within the people available and at or above its minimum staff. The case fixes how labor is costed: regular staff first, then cross-trained staff the partner department isn't using, then overtime, each at its own hourly rate including benefits. The case doesn't publish its scoring formula, so we treat anything beyond that order as an assumption.

> **Terms used throughout**
>
> **Block:** one warehouse, department, day and shift. The plan gives a total number of people for each block.
>
> **Utilization:** the hours of work a block's demand needs, divided by the hours its staff actually work. Above about 85-90%, service breaks down.
>
> **Meets target:** the block's wait or response time is at or under its department's target.
>
> **Chance of meeting target:** our model's estimate for one block, allowing for forecast error and callouts. **Expected service** is the average of these chances across open blocks.
>
> **Capacity:** the most people a block can field: its regular staff, plus usable cross-trained staff, plus overtime.
>
> **Draft:** the manager's draft schedule for the week, which the case provides. Appendix D has a full glossary.

2 The data

The case comes with six weeks of simulated history and a grid describing the planning week. Every number in this report comes from those files, through code that anyone can rerun.

Sources

Table 4. The data pack files we use. Every table has the same grain: one row per warehouse, department, date and shift.

File	Covers	Rows	What it holds
v_ShiftHistory.csv	Jan 1 to Feb 11, 2026	1,134	Demand, schedules, callouts, availability, cost and service for every block
v_PlanningWeekGrid.csv	Feb 12-18, 2026	567	Each scenario's availability, draft schedule, demand multipliers and pay rates. No demand
PlanningWeek_BenchmarkForecast.csv	Feb 12-18, 2026	63	The case's naive daily forecast, the floor to beat
TEAM_SUBMISSION_TEMPLATE.csv	Feb 12-18, 2026	567	The form we fill in
comp_LaborStandard.csv		3	Labor hours needed per workload unit, by department
comp_LaborCostAssumption.csv		9	Regular and overtime hourly rates

History against the planning week

- **The planning grid has no demand.** Forecasting it is part of the task, so chapter 4 builds that forecast.
- **Some columns exist only in the planning grid,** such as promotion strength and the stock-up and rebound flags. The forecast uses only what both periods share.
- **Holiday labels differ.** History names the MLK weekend and the planning grid names the Presidents' Day weekend. We map both to shared categories (the weekend before a holiday, and the holiday Monday), or no model could carry the effect across.
- **Closed blocks have no demand.** History has 81 closed blocks (New Year's Day and Pharmacy's closed shifts), leaving 1,053 open blocks. We leave closed blocks out of every fit.

Quality checks

We checked the files against each other and against the data pack's own documentation before using them. All of these passed:

- Every file has the expected rows and keys, and the template's keys match the planning grid exactly.
- Labor cost in history follows the documented formula exactly, and shift demand adds up exactly to daily demand.

- The benchmark is exactly what the docs say: the average of the same weekday over weeks 3 to 6, in the same units as history's demand. It is daily, so it must be split into shifts.
- The template's rows are sorted by shift name, not by time of day, so we always join on the keys and never on row order.

What the docs don't say

Reading the grid itself turned up several things the case documents leave out. Each one changes the plan.

- **DISRUPT cuts more than described.** It changes 36 blocks. Beyond the listed cuts, the partner department's cross-trained staff also falls: Warehouse 2 Merchandising on Feb 13-16, and Warehouse 4 Front End's opening shift on Feb 12-13. So we read availability for each scenario from the grid and never type in the described cuts.
- **The draft can't be staffed in DISRUPT.** In 12 blocks it schedules more people than are available, 36 people in all under the case's cost rule.
- **The draft is the same in every scenario.** It schedules the same 10,344 paid hours in BASE and PEAK. In DISRUPT, only 10,206.5 of them can actually be fielded. It ignores both the surge and the illness.
- **Promotions always fall on weekends and holidays.** In history, every Friday to Sunday and every holiday day had a promotion, and no other day did. Their effect can't be separated from the weekday and holiday effects, which already carry it.
- **The data pack's customer traffic index is an output, not an input.** Its own feature list marks it "never use as model input", because it is computed from the answer. Using it would leak the demand we are asked to forecast.

3 What history shows

Before planning anything, we asked what six weeks of history say about service, staffing and callouts. Four lessons shape the plan.

Service falls off a cliff

Over the six weeks, 77.9% of open blocks met their service target: 77.0% in Front End, 69.9% in Merchandising and 88.3% in Pharmacy. Whether a block met target depended almost entirely on its utilization (Figure 2). Below 84% utilization, 100% of blocks met target. Between 85% and 90%, only 62% did. At 90% or above, almost none did (0.6%).

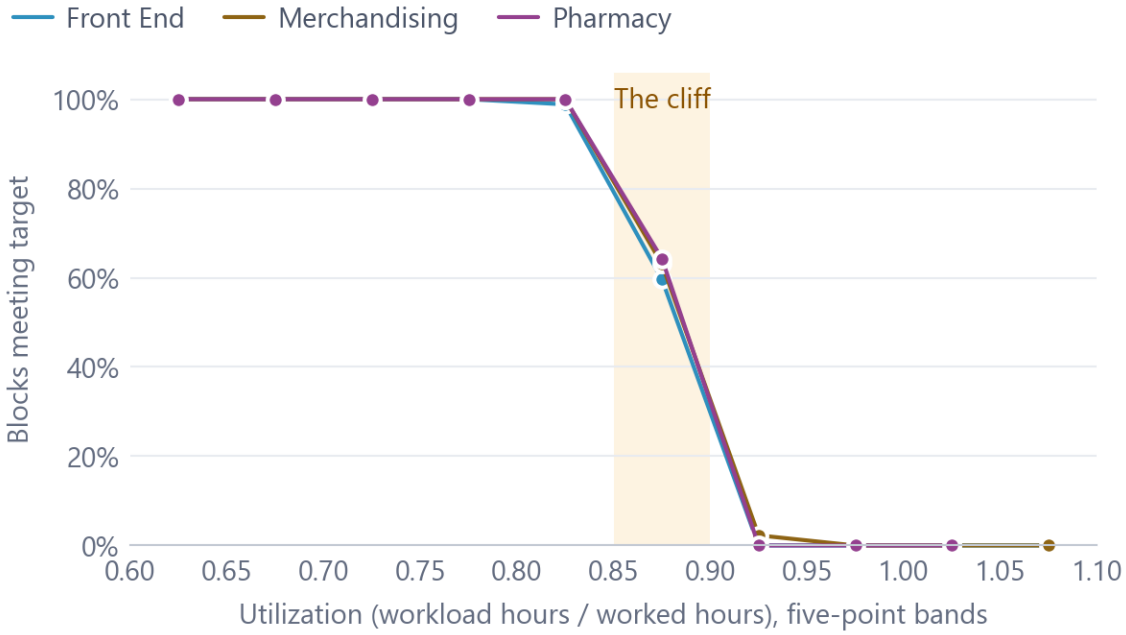


Figure 2. Share of historical blocks meeting target, by utilization, in five-point bands with at least three blocks. Past about 97% utilization, waits level off at roughly 4.6 times target in Front End, 4.1 in Merchandising and 4.3 in Pharmacy.

This shape drives everything else. A plan that aims right at 85% has no margin: a little extra demand, or one person calling out, pushes a block over the edge. What counts is the staff who actually work, not the staff on the schedule.

The right total hours, in the wrong places

Across all six weeks, staff worked 59,198 hours, within 1.2% of the 58,486 needed to run every block at 85% utilization, in whole people and never below minimum staff. So in total the warehouses had about the right hours. But 3,184 of those hours sat in blocks that didn't need them, worth $103,458 at regular rates, while 2,472 hours were missing from blocks that did.

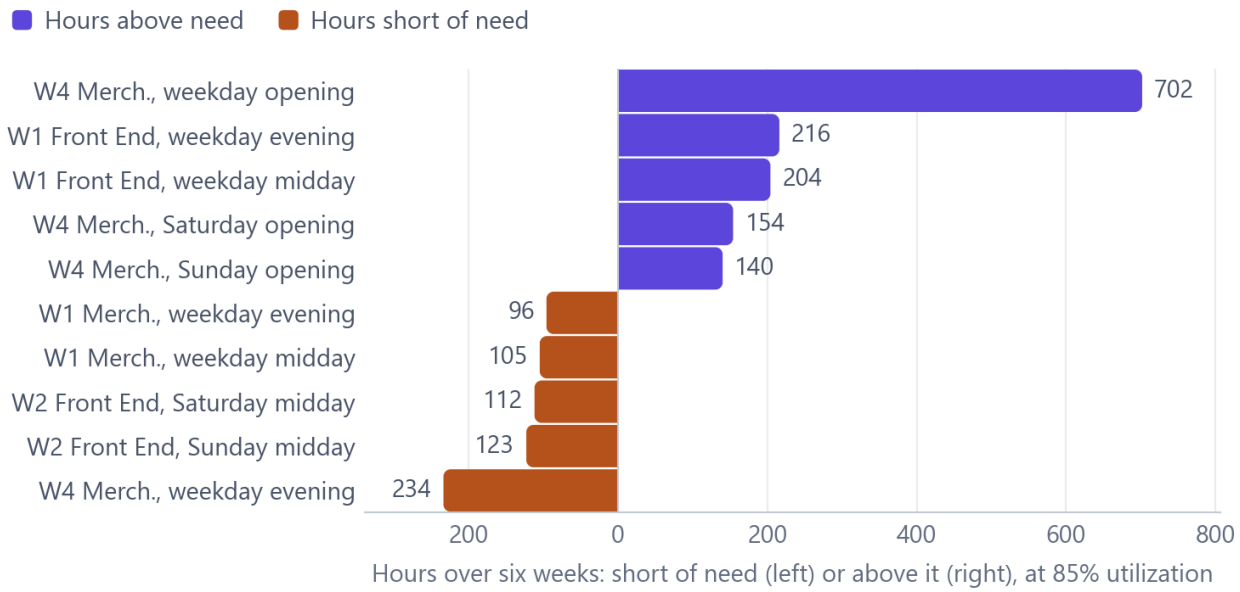


Figure 3. The block types with the largest gaps against need over six weeks, as worked. Iris bars had more hours than they needed; bronze bars had fewer. The full ranking is in outputs/diagnosis_staffing_gaps.csv.

Some gaps are clear habits. Warehouse 4 Merchandising's weekday opening crew ran at 67% utilization, while its weekday evening crew ran at 92% and met target only 28% of the time. That evening crew was 222 hours short before anyone called out. Other gaps are partly callouts: Warehouse 1 Merchandising's weekday evening gap falls from 96 hours as worked to 54 as scheduled.

Table 5. Six-week hours by department, against need at 85% utilization.

Department	Worked hours	Needed	Hours above need	Hours short	Value of the excess	Blocks missing target
Front End	26,856	26,574	1,284	1,002	$37,396	85
Merchandising	27,956	27,681	1,528	1,253	$46,860	111
Pharmacy	4,387	4,232	372	217	$19,203	37

Schedules barely follow demand. By department, weekly demand moved by up to 10% across the five full weeks, while scheduled hours moved by at most 2%. Managers schedule the same crews whatever the week brings. So moving hours comes before adding them.

Callouts are predictable

Overall, 5.2% of scheduled employees called out, and 49% of those were replaced with overtime. When a block's overtime was already used up, none were. Callouts follow a pattern: more on weekends, more on the day after a holiday, and many more during Warehouse 2's flu week in late January (Figure 4).

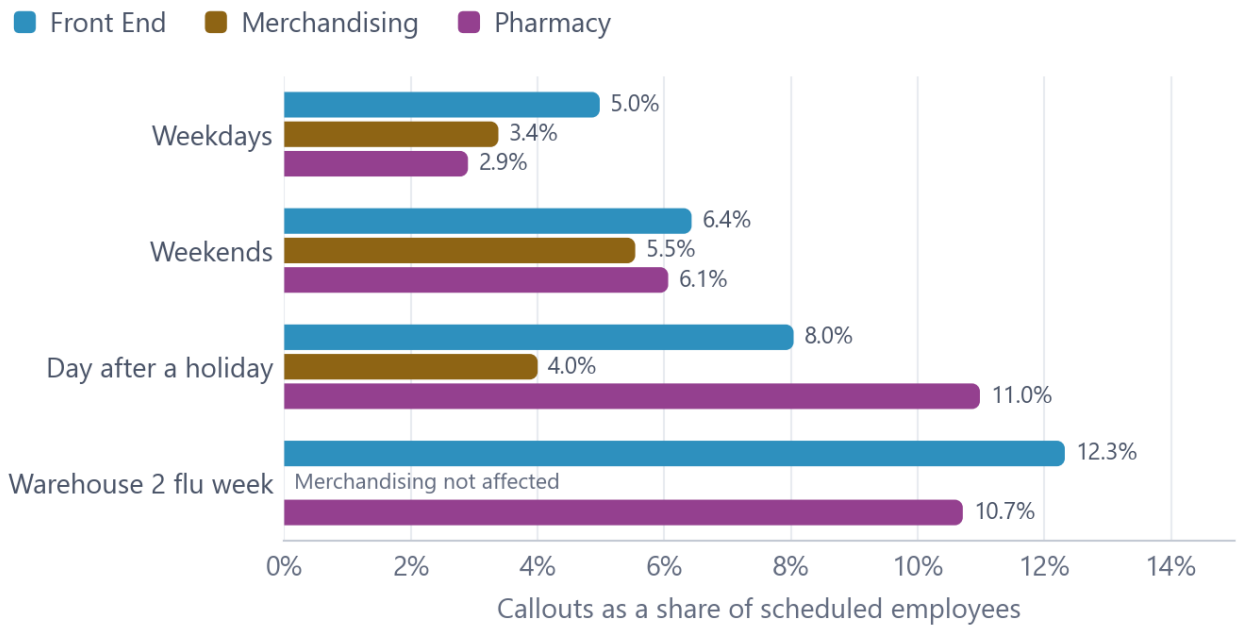


Figure 4. Callouts as a share of scheduled employees. The flu-week blocks (Warehouse 2 Front End and Pharmacy, late January) are kept apart from the other rows. The day-after-holiday rates rest on two days, and Pharmacy's rates on few people: 9 of 82 scheduled the day after a holiday.

Losing people hurts. Blocks that lost staff to callouts met target far less often: 65% against 87% in Front End, 47% against 77% in Merchandising, and 48% against 93% in Pharmacy. So the plan carries an allowance for callouts at these rates (chapter 5).

Overtime isn't the main cost lever

Labor cost over the six weeks was $1,890,242. Overtime was 3.5% of worked hours, and its premium over regular pay came to $21,345, or 1.1% of labor cost. Where the hours go matters far more than whether a few of them are overtime.

The draft schedule's weak points

The manager's draft for the planning week copies recent weeks: 99% of its open BASE blocks are within one person of the same weekday and shift's average over weeks 3 to 6. It is identical in all three scenarios. Under our forecast, 61% of its Front End and Merchandising blocks on the holiday weekend would run above 85% utilization in BASE, and 86% in PEAK. Our model expects only 68.5% of its BASE blocks to meet target.

4 Forecasting demand

The planning grid has no demand, so the first job is to forecast it: for every warehouse, department, day and shift. This chapter explains how, what we tested, and how far to trust the result.

The approach in plain words

We forecast each department's demand for each day, then split each day into its three shifts. The daily forecast starts from a normal day for that warehouse, department and weekday. It then applies a holiday effect where one applies, measured on the MLK weekend in January, the only holiday weekend in the history. Valentine's Day gets a small adjustment of its own. PEAK then multiplies the result by the case's surge factors.

The model

The daily forecast is a regression on the logarithm of demand, so every effect reads as a percentage ("the holiday weekend adds 12% to Front End"). It has four parts:

- **Warehouse level:** how busy each department is at each warehouse.
- **Weekday shape:** each department's own weekly pattern. Front End peaks on Saturday, while Merchandising peaks on Monday.
- **Holiday windows:** the weekend before a holiday and the holiday Monday, by department, plus the rebound after New Year's Day.
- **Weekly level:** this keeps a busy week from being mistaken for a holiday effect, since MLK week was the busiest of the six. A future week's level can't be known, so the forecast uses the average of the last 4 weeks.

Together these explain almost all of the variation in daily demand: warehouse and weekday alone reach 97.7%, holidays raise it to 99.5%, and weekly level to 99.9997%. We chose a regression over machine learning on purpose: there are only 1,053 open blocks of history and a single holiday weekend, too little to train a flexible model, and a regression's effects can be explained to a manager.

Drivers we tested

The case names several things that drive demand. The table shows what we tested for each, what we found, and how it enters the plan.

Table 6. The demand drivers the case names, and how each one is handled.

Driver	What we tested	What we found	How it enters the plan
Day of the week	Each department's weekday shape	Warehouse and weekday explain 97.7% of the variation	In the model
Holidays	New Year rebound, MLK weekend and MLK Day, by department	Holidays raise the fit to 99.5%	In the model, borrowing MLK's effects for Presidents' Day (A-05)
Weekly level	A level for each week	Weeks ran from 0.95 to 1.06 of average, with no trend	In the model, at the last 4 weeks' average (A-14)
Warehouse size	A level for each warehouse and department	Large warehouses are busier, as expected	In the model, and each warehouse gets its own service curve
Department activity	Labor hours per workload unit, from the case	Departments differ in units and in effort per unit	Converts demand into hours of work (chapter 5)
Weather (temperature)	Correlation with what the model leaves unexplained	−0.006: no effect. Front End's raw −0.57 is a calendar artifact	Left out. PEAK's storm comes in through the case's multipliers
Promotions	Correlation with what the model leaves unexplained	−0.017: promotions always fall on weekends and holidays, which already carry their effect	Left out. PEAK's promotion comes in through the case's multipliers
Customer traffic	The data pack's traffic index	An output of the simulation, marked never to use as an input	Not used: it would leak the answer
Valentine's Day	No history: it falls outside the six weeks	A judgment, tested for sensitivity	Adds 3% to Front End and Merchandising on Feb 14 (A-06)
Post-holiday rebound	Jan 20, flagged like Feb 17	Ran only 0.1% above its own week	Left out

Holiday effects

With weekly level in the model, the holiday effects match the factors in the data pack's own build script exactly, so the MLK effects themselves are measured, not guessed (Figure 5). Front End and Merchandising get busier over a holiday weekend. Pharmacy gets quieter, and on the holiday Monday its demand falls by 30%.

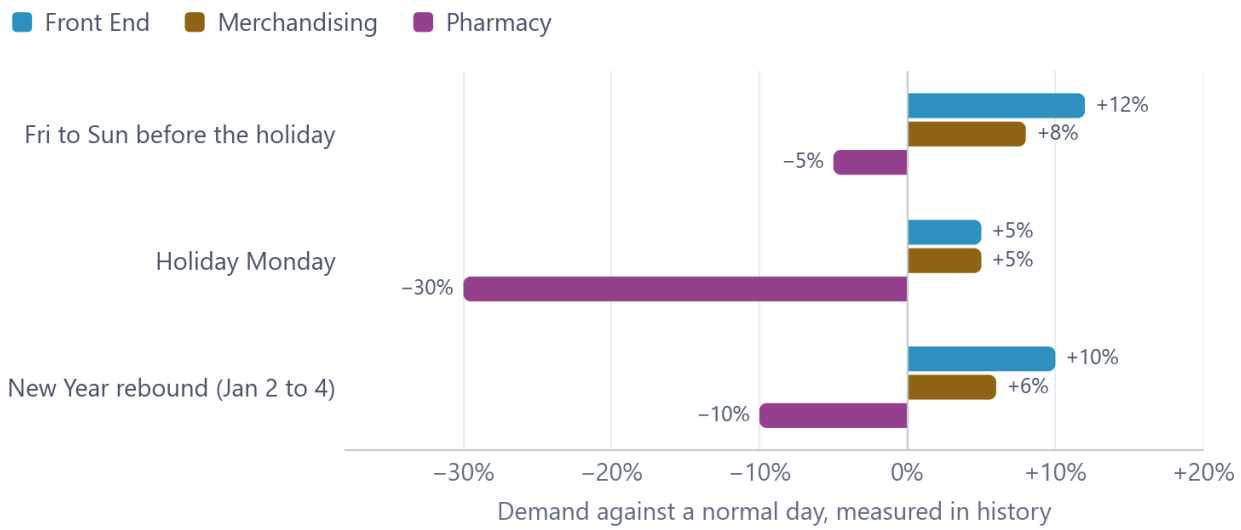


Figure 5. Holiday effects measured in history, as a change against a normal day. The MLK window's effects are applied to Presidents' Day.

Whether Presidents' Day repeats MLK is the assumption (A-05). We can test borrowing a holiday only once: forecasting the MLK weekend from the first two weeks, using the New Year rebound as the guide. Our error was 9.6% against the benchmark's 10.2%: better for Front End (8.2% against 12.4%) and Merchandising (8.7% against 11.3%), worse for Pharmacy (11.9% against 7.0%). So the plan carries extra forecast uncertainty on Feb 13-16: 2% for Front End and Merchandising, and 5% for Pharmacy.

Valentine's Day is a judgment

History holds no Valentine's Day, and the data pack gives Feb 14 the same features as the days around it. The case still asks us to forecast it, so we add 3% for Front End and Merchandising on that Saturday, on top of the holiday weekend, and nothing for Pharmacy (A-06). The table shows how little the answer depends on that choice.

Table 7. Feb 14 Front End and Merchandising blocks in BASE. Our plan costs $59,195 for those blocks.

If the true uplift is	Expected service, our plan	Had we planned for it	Feb 14 cost, had we planned for it
0%	91.0%	90.3%	$57,868
3%	85.3%	85.3%	$59,195
5%	80.4%	80.7%	$60,096
10%	64.5%	65.9%	$60,459

Planning for a bigger uplift barely helps, because 11 of those 18 blocks are already at capacity. The size of this assumption matters less than the shortage of people that day.

The shift split

Each day's forecast is split into opening, midday and evening using each department's average shares for that kind of day. Front End's demand moves about 3 points toward midday on holiday-window days, so Front End gets its own holiday shares. Leaving one week out, that cut the split's error on Front End holiday days from 5.6% to 4.2%. Other departments didn't improve, so they keep one set of shares.

Accuracy, and its limits

We tested the forecast the honest way: forecast each of weeks 4, 5 and 6 using only the weeks before it, and compare with what happened. The benchmark's method gets the same test.

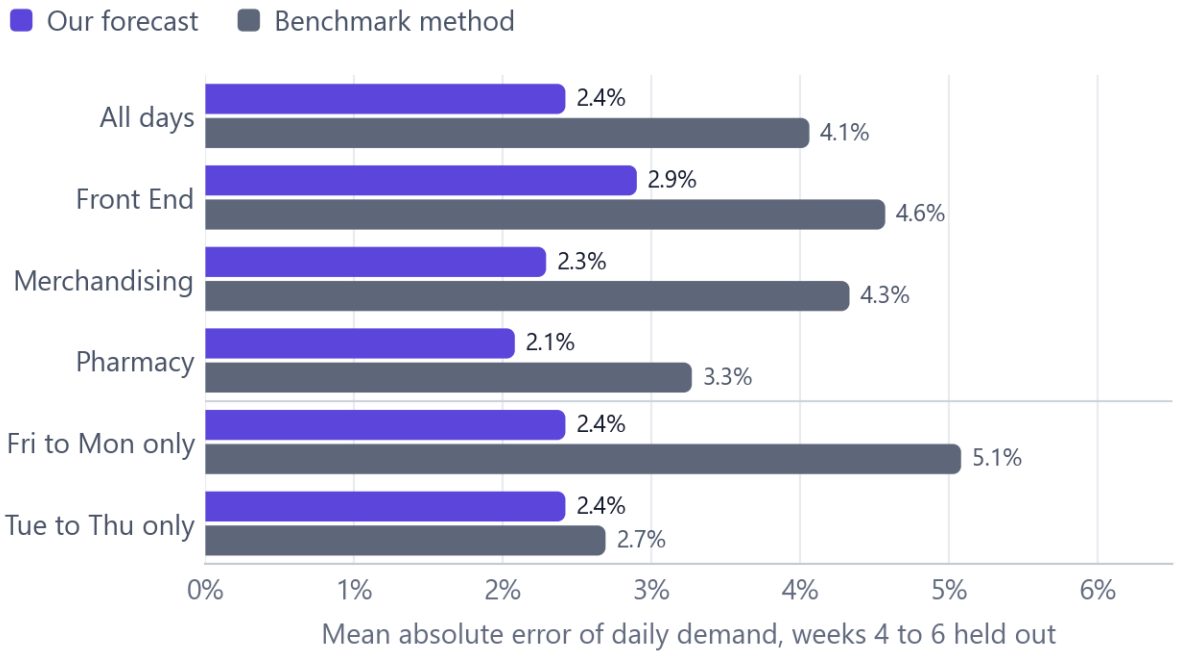


Figure 6. Mean absolute percentage error of daily demand on held-out weeks. By week, ours against the benchmark's: 1.6% against 4.8%, 4.2% against 5.5%, and 1.4% against 1.8%. By block, after the shift split: 4.2% against 5.4%.

Our daily error was 2.4% against the benchmark method's 4.1%. **But the edge comes entirely from Friday to Monday**, where the benchmark's four-week average still held earlier holiday days: 2.4% against 5.1%. From Tuesday to Thursday the two nearly tie (2.4% against 2.7%). So the test shows that our forecast removes holiday distortion. It doesn't show that we beat the benchmark in the planning week: these held-out weeks hold no holidays, and three weeks is a small test.

The planning-week forecast

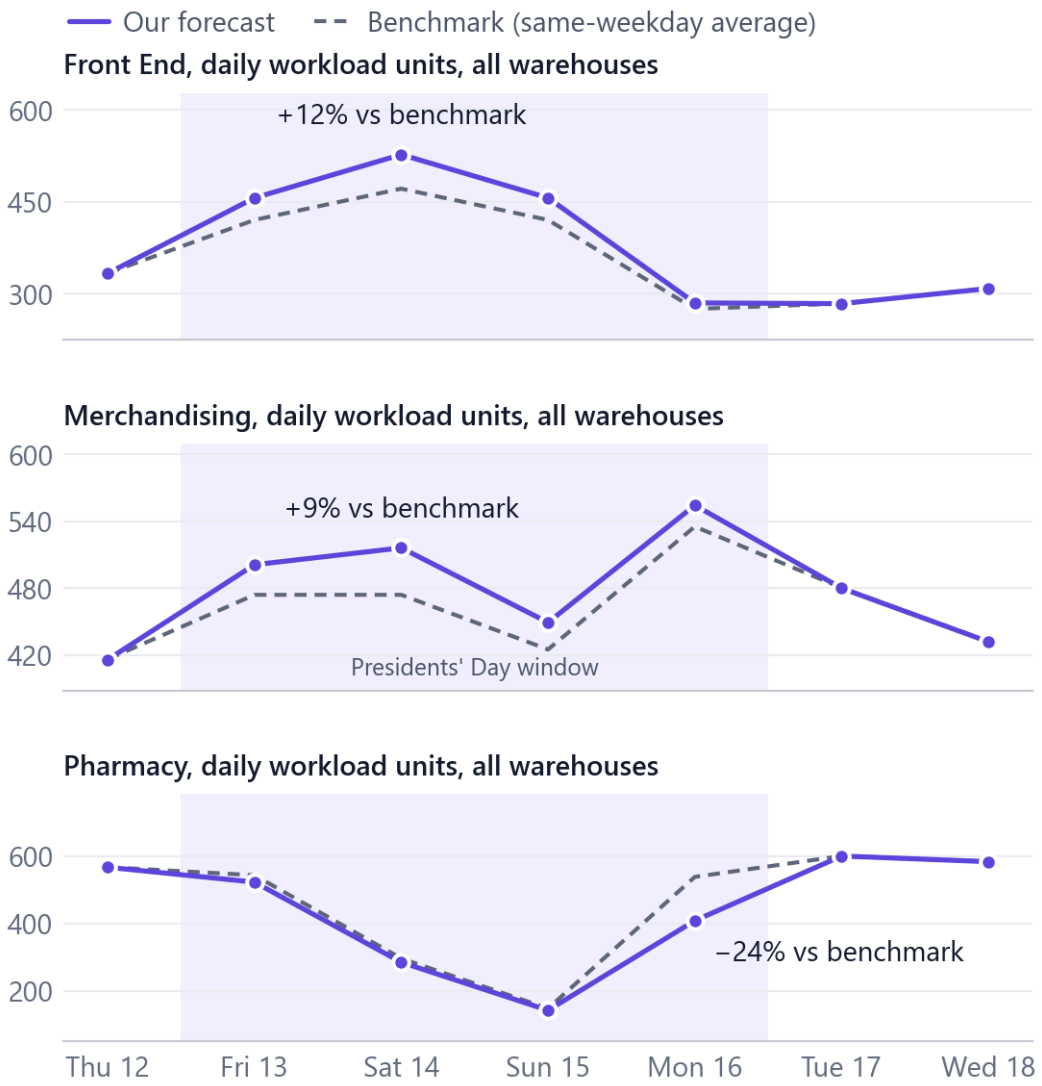


Figure 7. Our BASE daily forecast against the benchmark, summed across warehouses. On normal days the two agree within 0.1%, because both start from the same recent weeks. They differ only in the Presidents' Day window.

The benchmark averages four weeks, only one of which held the MLK weekend, so it carries just a quarter of the holiday effect. That is why it runs low for Front End on Valentine's Day (our forecast is 11.8% above it) and high for Pharmacy on Presidents' Day (ours is 24.1% below it). Over the whole week, our Front End forecast is +5.4% against the benchmark, and Pharmacy's is −5.1%.

Forecast error still matters, so the risk model carries it: the held-out error per block was 5.6% for Front End, 5.3% for Merchandising and 4.5% for Pharmacy, plus the extra holiday uncertainty above.

5 From demand to staff

A forecast says how much work is coming. This chapter turns it into a number of people for every block: how service responds to workload, how much risk to allow for, how staff are costed, and how the plan picks the best staffing in each shift.

The service curve

For each department, we fitted a curve to every historical block: how far its wait or response time ran above or below target at each level of utilization. The curve rises slowly, bends sharply upward near the cliff, then flattens once waits stop getting worse. Each warehouse gets its own starting level, because they break at slightly different points. Outcomes scatter around the curve, and the size of that scatter turns the curve into a chance of meeting target.

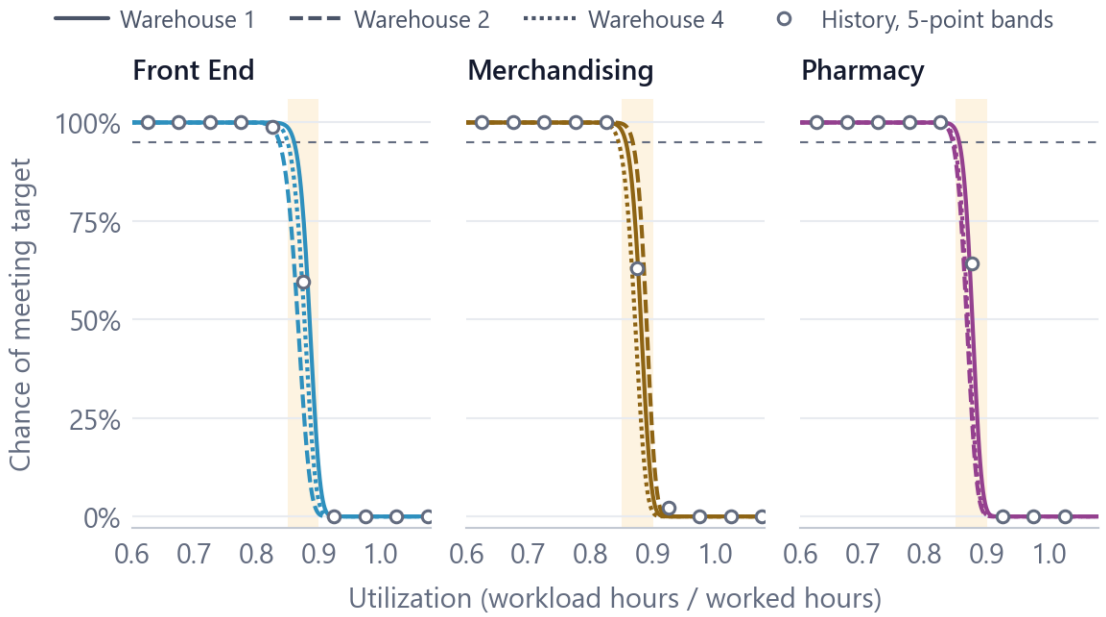


Figure 8. The fitted chance of meeting target at each utilization, one line per warehouse, with history's observed share in five-point bands. The dashed line is the service goal, and the shaded band is the cliff, 85-90% utilization.

Using each block's measured wait, rather than only whether it passed, matters because few blocks sit right at the cliff. Our first curve bent at a sharp corner. A calibration study found it slightly optimistic near the goal, mostly in Front End (F-62), and showed that a smooth bend fits better on every measure (F-64). We switched to the smooth bend before freezing the numbers (D-29). That moved the service limits below by less than one point of utilization and added one person to some blocks.

Table 8. Utilization at which a block has a 95% chance of meeting target, before allowing for forecast error and callouts.

Department	Warehouse 1	Warehouse 2	Warehouse 4
Front End	86.1%	83.7%	85.0%
Merchandising	85.7%	86.8%	84.5%
Pharmacy	85.5%	84.5%	84.7%

So a high chance of meeting target needs utilization at or below about 84% to 87%, depending on the department and warehouse. **85% utilization is a ceiling, not a target (A-04).** The starter queries in the data pack aim right at it, which leaves no room for anything to go wrong.

Allowing for risk

Three things can push a block over the cliff on the day: demand above forecast, callouts that aren't replaced, and the scatter around the curve. For each block and each possible headcount, the risk model adds up all three. Forecast error comes from the held-out tests in chapter 4. Callouts come at the historical rate for the department and situation: weekday, weekend, the day after a holiday, or illness. About 49% of callouts are replaced with overtime, but only while the block has overtime room left. A simulation of 200,000 random draws checks the calculation, and a test keeps the two within half a point.

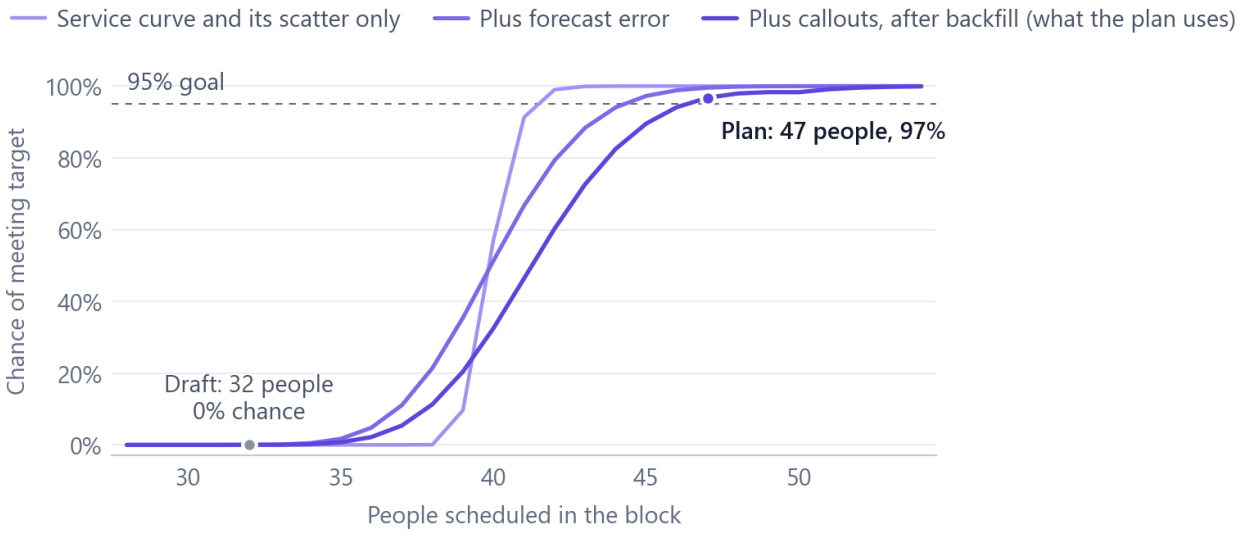


Figure 9. One block's chance of meeting target at each headcount: Warehouse 2 Front End, Sun Feb 15, midday, in BASE. Each line adds one more risk. The plan takes the smallest headcount whose chance on the darkest line reaches the goal.

With the draft's 32 people, this block would run at 108% utilization and almost certainly miss its target. Looking only at the curve and its scatter, 42 people would reach the goal. Allowing for forecast error raises that to 45, and allowing for callouts to 47. The plan schedules 47, for a 97% chance at 73% utilization. Ignoring risk would leave this block on the edge of the cliff.

Allocating and costing staff

The case fixes how a block's people are filled and paid. Regular staff come first. Then come cross-trained staff from the partner department, but only those the partner isn't already using in that shift. Overtime comes last. Regular and cross-trained staff are paid the regular hourly rate, and overtime the overtime rate. A block can't field more than its capacity, and a plan that tries is infeasible.

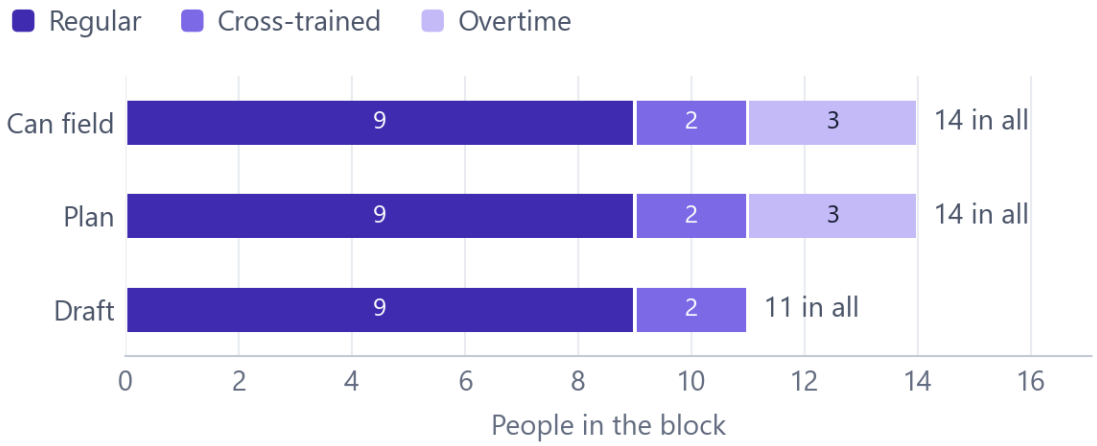


Figure 10. How one block is filled: Warehouse 1 Merchandising, Mon Feb 16, midday, in BASE. The plan uses every person the block can field.

Table 9. The cost of that block under the case's rule. Its partner, Front End, plans 18 of its 20 regular staff in the same shift, so 2 are free to cross over.

Source	People	Paid hours	Hourly rate	Cost
Regular staff	9	45	$30.00	$1,350
Cross-trained, from Front End	2	10	$30.00	$300
Overtime	3	15	$39.60	$594
Block total	**14**	**70**		**$2,244**

The search

For each block, the search looks for the smallest headcount that reaches a 95% chance of meeting target. Front End and Merchandising share cross-trained staff in the same shift, so the search solves the two together. In each shift it tries every feasible pair of headcounts, applies the cost rule to both, and keeps the best pair in this order:

1. **Service first:** the highest total chance of meeting target across the two departments, counting no block above the goal.
2. **Then shortfall:** the fewest hours short of what the goal would need, where capacity runs out.
3. **Then cost:** the cheapest pair.

Pharmacy has no cross-trained staff, so it is solved on its own. A department only lends people it doesn't need itself (D-17); without that rule, the search cut the busiest blocks to lend their staff, which no warehouse leader would accept. With 63 shifts per scenario, trying every pair takes seconds, so no optimization solver is needed, and the answer is exact under the case's own cost rule and the real service curve (D-06).

In the example shift, Front End needs only 18 people for a 98% chance, which frees 2 cross-trained staff. Merchandising takes both and fills its last 3 places with overtime, for a 99% chance. Without the cross-trained staff, it would need 2 more overtime places it doesn't have.

Why a total per block, not individual shifts

- **The case asks for totals.** The submission template has one row per warehouse, department, day and shift, and asks for the total number of employees in each. That is the decision the plan answers.
- **The data has no one to name.** Availability comes as counts per block: regular staff, overtime and cross-trained staff. There are no individual employee records, skills or preferences to build personal schedules from.
- **Managers can fill the totals from their own roster.** Each block's total comes with its split into regular, cross-trained and overtime staff, in the order the case's rule fills them.

Choosing the 95% goal

The goal is one setting, and the plan can be rebuilt at any other in minutes. Raising it buys service at a rising price (Figure 11). Going from 92.5% to 95% adds $7,828 for 1.2 points of expected service. Going from 97.5% to 99% adds $9,926 for just 0.4 points. Service is judged first, so we chose 95%: high enough to protect service, below the point where each extra point gets very expensive (A-13).

Table 10. Cost and expected service in BASE at each service goal. More blocks fall short of a higher goal, because capacity stays fixed.

Service goal	Labor cost	Against the draft	Expected service	Blocks below the goal
85%	$355,885	+8.1%	90.9%	11
90%	$364,289	+10.7%	93.4%	14
92.5%	$369,677	+12.3%	94.4%	21
95% (chosen)	$377,504	+14.7%	95.6%	25
97.5%	$388,724	+18.1%	96.4%	36
99%	$398,650	+21.1%	96.8%	56

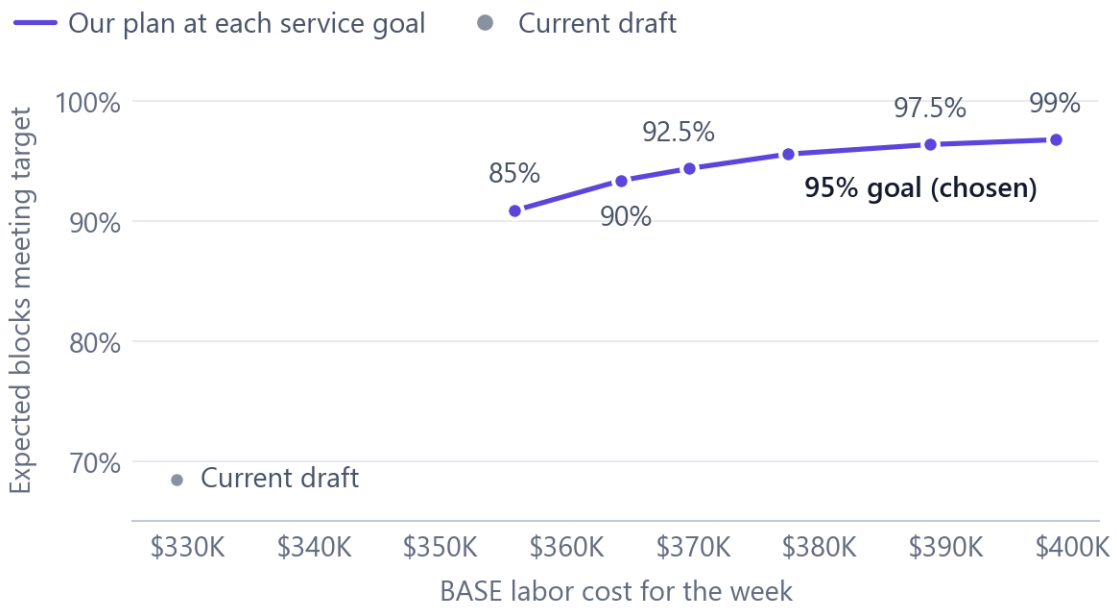


Figure 11. BASE labor cost against expected service at each service goal, with the draft for comparison. Each extra point of service costs more than the last.

What the allowance for callouts costs

The case doesn't say whether judges apply callouts to submitted staffing. So we built three plans, each allowing for a different set of risks, and scored each one under three sets of conditions.

Table 11. Three BASE plans, scored under each set of conditions. Service is expected service across open blocks.

Plan allows for	Labor cost	If both risks happen	Forecast error, no callouts	Neither
Both risks (our plan)	$377,504	95.6%	99.6%	100.0%
Forecast error only	$354,761	90.4%	98.3%	100.0%
Neither	$328,220	73.4%	86.2%	99.4%
The current draft	$329,108	68.5%	76.9%	80.1%

The "Neither" row is a finding of its own. Just moving hours to where they are needed, at about the draft's cost, beats the draft under every set of conditions. The allowance for callouts costs $22,743. If judges ignore callouts, it gains little (98.3% to 99.6%). If they apply them, it lifts expected service from 90.4% to 95.6%. Service is judged first and history shows callouts are real, so we plan for them (A-03).

6 Results

This chapter compares the plan with the manager's draft in every scenario, by department and by day, and shows where even the best plan runs out of people.

Each scenario against the draft

Table 12. The plan against the draft in each scenario, for all three warehouses. Both are scored by the same risk model. Costs follow the case's rule and count only staff who can be fielded.

	BASE	PEAK	DISRUPT
Draft labor cost	$329,108	$329,108	$326,367
Plan labor cost	$377,504	$387,926	$366,649
Change in cost	+14.7%	+17.9%	+12.3%
Draft paid hours	10,344.0	10,344.0	10,206.5
Plan paid hours	11,657.5	11,923.0	11,288.0
Plan overtime hours	974.5	1,224.0	1,014.0
Draft expected service	68.5%	58.3%	66.6%
Plan expected service	**95.6%**	**87.0%**	**89.5%**
Draft blocks over capacity	0	0	12
Plan blocks over capacity	0	0	0
Plan blocks below the 95% goal	25	45	39

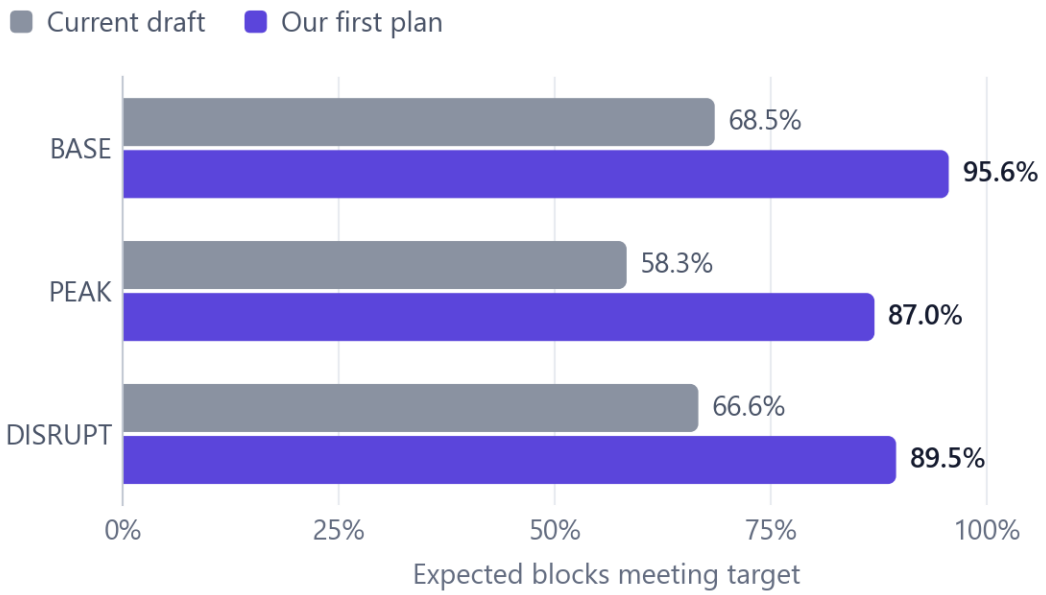


Figure 12. Expected share of blocks meeting target, draft against plan. The same values are in Table 12.

BASE is the expected week. The plan adds 1,314 paid hours, mostly on the holiday weekend, and lifts expected service by 27.1 points.

PEAK is the hardest week. The plan adds 266 hours against BASE, but the surge lands on the days when holiday time off already leaves the fewest people, so expected service reaches only 87.0%. That is still far above the draft's 58.3%.

DISRUPT has fewer people, not more demand. The plan uses 370 fewer hours than BASE because they aren't available, and unlike the draft it never schedules people who can't be fielded. The draft is over capacity in 12 blocks; the plan in none.

By department, in every scenario

Table 13. Hours, cost and expected service by department, all warehouses. The full block-by-block comparison is in outputs/plan_vs_draft.csv.

Scenario	Department	Draft hours	Plan hours	Draft cost	Plan cost	Draft service	Plan service
BASE	Front End	4,729	5,410	$139,230	$163,346	61.9%	95.2%
	Merchandising	4,846	5,404	$149,673	$169,456	58.5%	95.0%
	Pharmacy	770	844	$40,206	$44,703	87.7%	96.6%
PEAK	Front End	4,729	5,494	$139,230	$166,841	49.1%	79.4%
	Merchandising	4,846	5,586	$149,673	$176,383	43.8%	86.7%
	Pharmacy	770	844	$40,206	$44,703	86.0%	96.1%
DISRUPT	Front End	4,623	5,065	$137,027	$152,676	58.5%	80.2%
	Merchandising	4,834	5,392	$150,024	$169,729	58.3%	94.5%
	Pharmacy	750	832	$39,316	$44,244	85.6%	94.5%

Pharmacy changes least, because it needs few people and history ran it close to need. Front End takes the biggest hit in both PEAK and DISRUPT: its expected service falls to 79.4% and 80.2%, because the surge and the illness both land on it.

By day

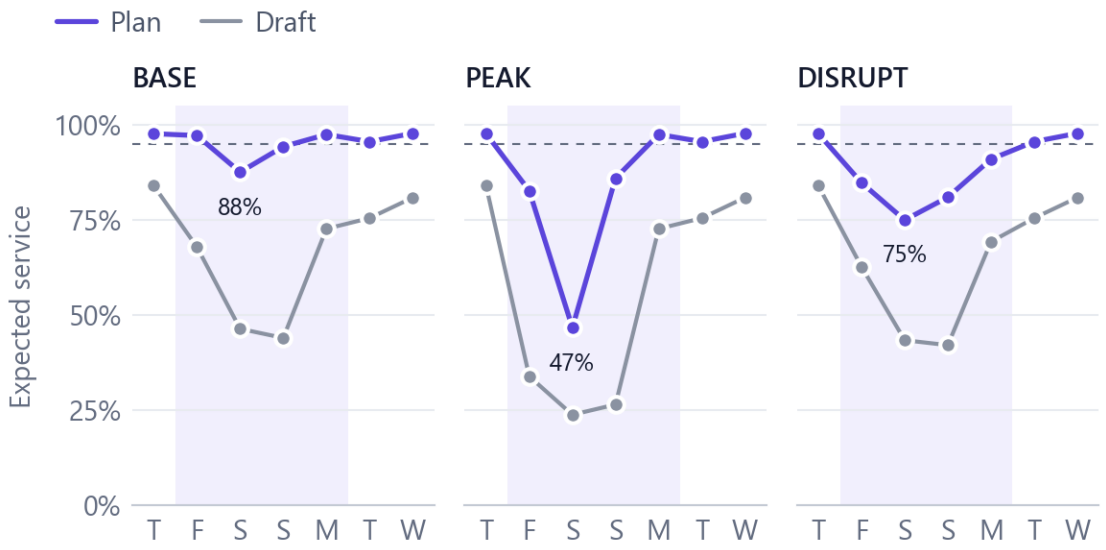


Figure 13. Expected service by day, draft against plan, in each scenario. The dashed line is the goal and the shaded days are the Presidents' Day window. The label marks the plan's lowest day.

All three scenarios dip on the holiday weekend, and most on **Saturday, Feb 14**: Valentine's Day, the busiest day of the window, and a day when holiday time off leaves fewer people. In BASE, the plan uses 98% of the hours the warehouses can field that day, against 65% on Thursday, Feb 12 (Figure 14).

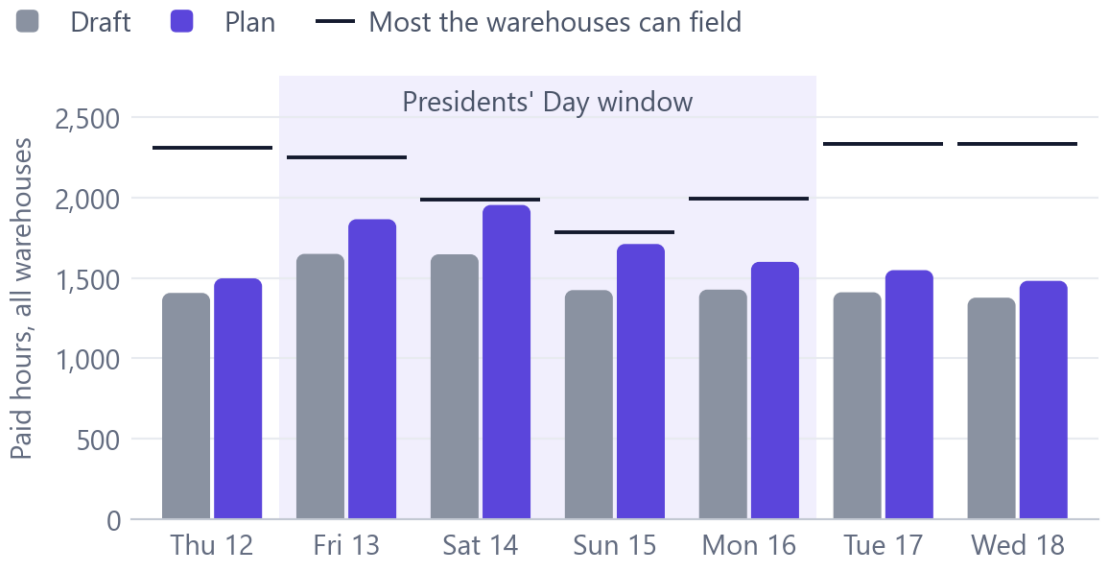


Figure 14. BASE paid hours by day, draft against plan, with the most the warehouses could field each day (regular staff, usable cross-trained staff and overtime).

Where capacity runs out

Some blocks can't reach the goal even using every regular and overtime person they have. Table 14 counts the extra people they would need.

Table 14. People short of the 95% goal, summed over the blocks below it. Every one of these blocks already uses all its own regular staff and overtime.

Day	BASE	PEAK	DISRUPT
Fri Feb 13	4	42	44
Sat Feb 14	25	113	79
Sun Feb 15	9	30	56
Mon Feb 16	1	1	17
Tue Feb 17	4	4	4
People short, all week	**43**	**190**	**200**
Blocks below the goal	25	45	39

Being below the goal doesn't mean a block is expected to fail. In BASE, the 25 blocks below the goal have an average chance of 83%, and 11 of them sit between 90% and the goal. Of these, 22 fall on Feb 13-16, including 11 on Feb 14. On that day, 11 of the 18 Front End and Merchandising blocks are at capacity. This is an availability problem: rearranging the schedule can't fix it, but changing who is available can (chapter 7).

The Feb 17 shortfall rests on thin data. The 3 BASE blocks short that day are all Pharmacy, driven by its 11% day-after-holiday callout rate, which rests on few people. At Pharmacy's 2.9% weekday rate, only 1 of them would stay below the goal (A-15).

Where the plan adds and where it cuts

In BASE, the plan adds people to 132 open blocks, cuts 12 and leaves 36 as the draft has them. Cuts come where history shows the draft's habit of scheduling more people than a block needs (chapter 3). Every change comes with its reason, filled in from the block's own numbers. Three examples from BASE:

> **Added: Warehouse 2 Front End, Sun Feb 15, midday**
>
> **Add 15: 32 in the draft to 47 in the plan.** The draft's 32 people would run at 108% utilization, with a 0% chance of meeting target. Adding 15 brings it to 73% utilization and a 97% chance. The weekend before Presidents' Day: 12% more Front End demand than a normal day, going by the MLK weekend. Before a miss becomes the likelier outcome, this block can lose 7 people, or take 18% more demand.

Cut: Warehouse 4 Merchandising, Wed Feb 18, opening

Cut 3: 21 in the draft to 18 in the plan. The draft's 21 people would run at 64% utilization, with a 100% chance of meeting target. That is more than this block needs. With 18, it runs at 75% utilization and still has a 98% chance. A normal Wednesday for Merchandising. Before a miss becomes the likelier outcome, this block can lose 2 people, or take 17% more demand.

Short of the goal: Warehouse 1 Front End, Sat Feb 14, midday

Add 5: 29 in the draft to 34 in the plan. This block can't reach the goal. It needs 39 people for a 95% chance of meeting target, but the plan can only field 34, using all of its own regular staff and overtime. The draft's 29 people would run at 96% utilization, with a 5% chance of meeting target. The plan's chance is 57%. The weekend before Presidents' Day: 12% more Front End demand than a normal day, going by the MLK weekend. Valentine's Day adds 3%, as a judgment call.

The reasons for all blocks in every scenario are in outputs/explanations.csv, and the tool shows each one when a block is clicked.

7 What a warehouse leader should do

A plan is only useful if someone can act on it. This chapter turns the results into actions: one before the week, one for choosing the plan, and rules for the day itself.

Before the week: limit holiday PTO

Holiday-weekend time off is the main reason Feb 14-15 run short. Front End's available regular staff falls by about 4 people per block that weekend, compared with recent weekends. We reran the BASE search keeping some of those people working, never more than each block lost against the last three normal weeks.

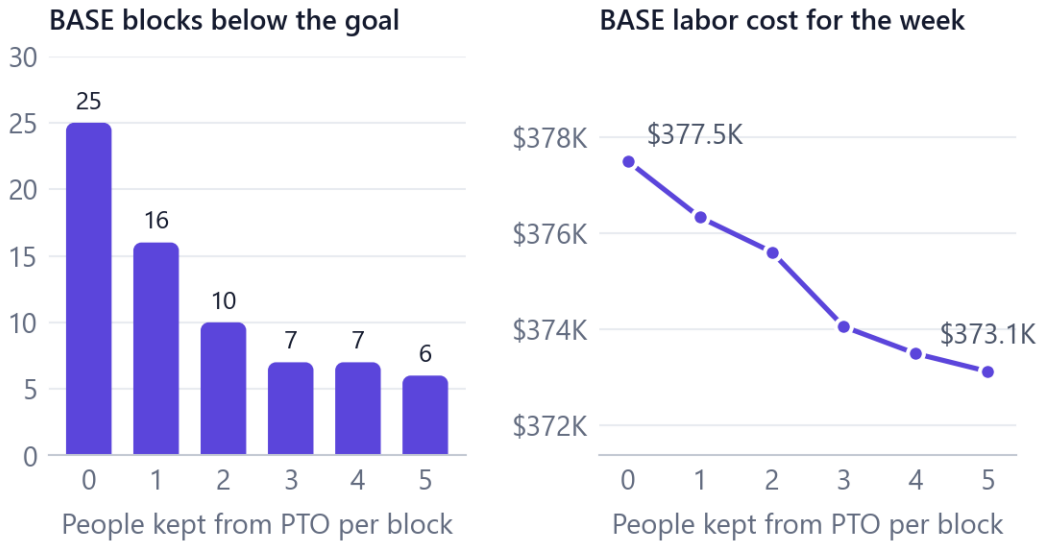


Figure 15. The holiday-PTO test in BASE: blocks below the goal, and labor cost, as more people per block are kept from holiday time off.

Table 15. Each row reruns the BASE search with up to that many people per block kept from holiday PTO, at the regular rate.

People kept per block	Employee shifts kept	Blocks below the goal	Expected service	Labor cost	Overtime hours
None (the plan)	0	25	95.6%	$377,504	974.5
1	62	16	96.7%	$376,337	788.0
2	115	10	97.1%	$375,613	646.5
3	150	7	97.2%	$374,056	527.5
4	171	7	97.2%	$373,495	466.0
5	179	6	97.2%	$373,123	437.0

Service rises and cost falls, because regular staff replace overtime. Most of the gain comes from the first few people per block. At the highest limit, blocks below the goal fall from 25 to 6, and none of those left

falls on a day holiday PTO touches. **This is the strongest single action we found.** It assumes some PTO is still open to change, which the data can't show (A-16).

Which plan to run

Each scenario's plan is ready in advance. These are the signs that should trigger a switch, and what to do in each case.

BASE: Baseline planning week

Use it when: No promotion surge or illness has been confirmed. This is the default.

- Schedule the BASE plan: 11,658 hours, at a cost of $377,504.
- Approve 975 hours of overtime ahead of time, mostly for the holiday weekend, so it isn't arranged at the last minute.
- Deal with the 43-person gap across 25 blocks before the week starts. The largest is Warehouse 2 Front End on Sat Feb 14, at 6 people across its shifts. Limiting holiday PTO is the most direct fix.

PEAK: Promotion and winter-weather surge

Use it when: The Feb 13 member promotion is confirmed, or a winter-storm advisory covers Feb 13-14.

- Switch to the PEAK plan: +265 hours against BASE, at a cost of $387,926.
- Put extra staff first into Warehouse 4 Merchandising, which gains the most (12 more employee shifts across the week).
- Approve 1,224 hours of overtime, against 975 in BASE.
- Expect waits even with this plan. It is 190 people short across 45 blocks, led by Warehouse 4 Front End on Sat Feb 14 (26 people across its shifts). Arrange outside cover, or tell members and staff what to expect.

DISRUPT: Callouts and reduced availability

Use it when: Warehouse 2 reports the illness cluster for Feb 13-16 (about 30% of Front End unavailable, one Pharmacy team member out per block), or Warehouse 4's opening stocking crew is about 35% short on Feb 12-13.

- Switch to the DISRUPT plan: -370 hours against BASE, because fewer people are available, at a cost of $366,649.
- Don't use the current draft here: it schedules 36 more people than can be fielded, across 12 blocks.
- Warehouse 2 Front End loses the most (100 fewer employee shifts across the week). Move every available cross-trained person there once Merchandising's own need is met.
- Escalate early. The plan is 200 people short across 39 blocks, led by Warehouse 2 Front End on Sat Feb 14 (57 people across its shifts), and no schedule can close that gap.

On the day

Each block has a **cushion**: the number of people it can lose, or the extra demand it can take, before missing target becomes the likelier outcome. The tool shows each block's cushion. These rules use it.

Table 16. Rules for the day, from the playbook.

If	Then
Callouts in a block pass its cushion. The typical cushion in people is: Front End 3, Merchandising 2, Pharmacy 1.	Replace them in this order: cross-trained staff if the partner department has people to spare, since they cost the regular rate, then overtime if the block has room. Each block's own cushion shows when you open that block in the planner.
Traffic runs above forecast by more than the cushion. The typical cushion is: Front End 20%, Merchandising 19%, Pharmacy 42%.	Treat the rest of the day as PEAK: bring in the extra staff for later shifts, starting with the blocks closest to their limit.
Waits pass target: 5 minutes at checkout, 60 minutes for replenishment, or 20 minutes in Pharmacy.	Don't wait for the next shift. Service falls off a cliff near 85-90% utilization, so one late person can make a block miss. Pull in help now.
Traffic runs well below forecast, with no callouts.	Offer voluntary time off in overtime blocks first, because overtime costs about 32% more per hour. Don't cut below minimum staff.

8 The Workforce Planner tool

The case asks for more than a dashboard: a working tool a warehouse leader could use. The Workforce Planner takes a scenario and shows staffing, labor cost and service risk, explains every recommendation, and re-plans live when conditions change.

What it's for

A warehouse leader uses it to answer five questions: what should we schedule this week, what will it cost, where are we at risk, why did each block change, and what should we do if conditions shift. Every number comes from the same engine that produced this report and the submission file, so the tool, the report and the submission always agree.

How it's built

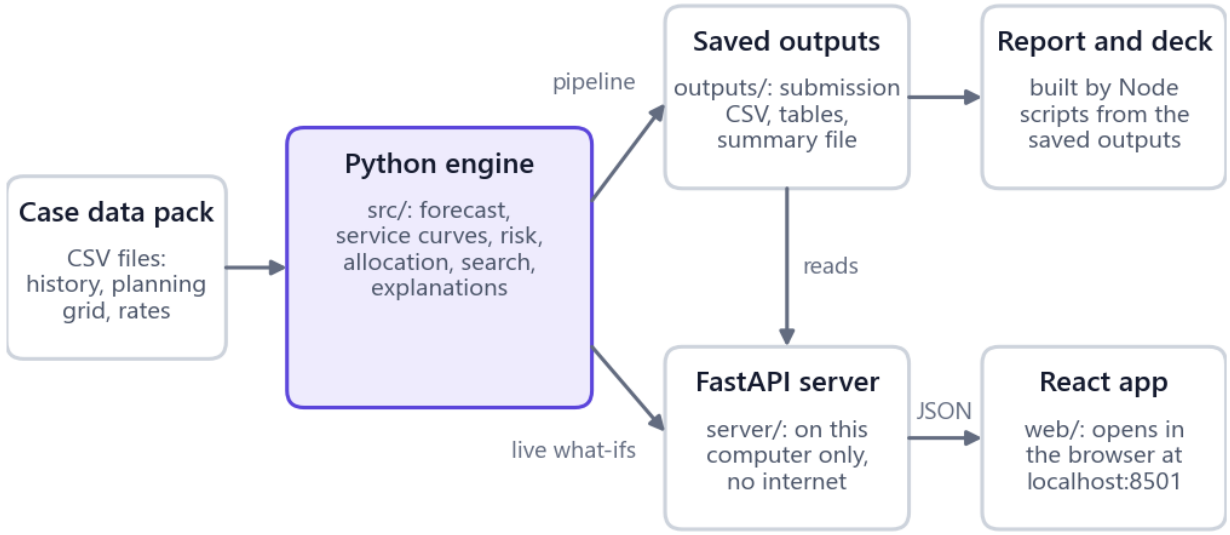


Figure 16. How the pieces fit. The engine runs once to write the saved outputs, and again on demand for live what-ifs. Everything runs on one laptop.

The engine is the Python code that does the analysis: the forecast, service curves, risk model, cost rule, search and explanations. A small FastAPI server wraps it in a web interface that the React front end calls. The server listens only on the laptop itself, and nothing loads from the internet, so the demo works with Wi-Fi off. We chose this over a dashboard tool because the analysis stays in tested Python, and the interface is a real web front end, the way such a tool would be deployed (D-24). An earlier Streamlit version remains as a fallback.

How to start it

The project lives in a private GitHub repository. A teammate with access needs Python 3.13 and runs these commands from the cloned folder. Running the app needs no Node.js, because the built front end is included. The full steps are in README.md.

```
git clone https://github.com/cgonztx-gif/misa-workforce-planner.git
cd misa-workforce-planner
python -m venv .venv
.venv\Scripts\activate          # Windows (Mac or Linux: source .venv/bin/activate)
python -m pip install -r requirements.txt
python -m zipfile -e MISA.zip .
python -m zipfile -e "MISA/Student Teams/Costco_Case_Competition_TEAMS.zip" data
python -m server                 # opens http://localhost:8501 when ready
```

On Windows, double-clicking start_app.bat does the last step and opens the browser. The server prepares all three scenarios when it starts, which takes 15-20 seconds behind a "preparing" screen. After that, every page and scenario switch is instant. **Before a demo, start the app a minute early.**

A guided tour

The dark top bar holds the two choices that apply everywhere: the **scenario** (BASE, PEAK or DISRUPT) and the **warehouse** (all, or WH1, WH2 or WH4, each with its size). They carry across pages, and the address keeps them, so a refresh keeps the view. A "Simulated data" badge sits beside the logo on every page. The sidebar groups the pages into the plan, what to do when things change, and how it works, and its Download CSV button saves the submission from anywhere.

Overview: the answer

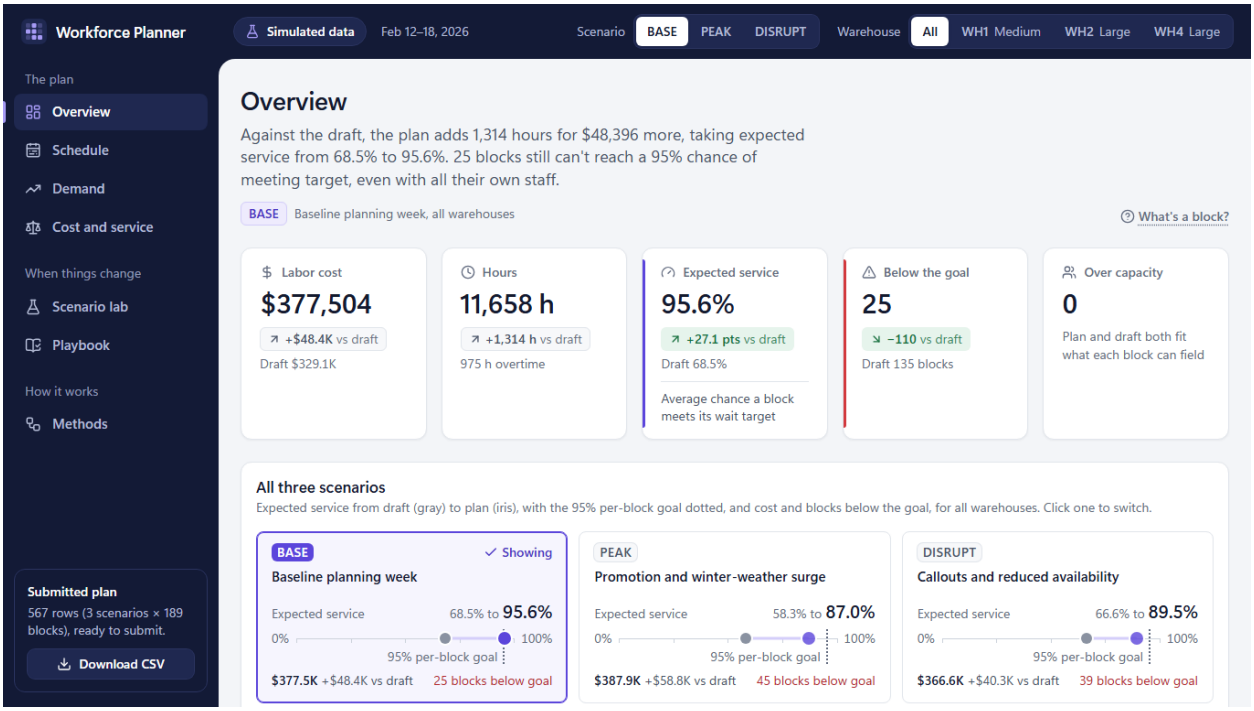


Figure 17. The Overview page for BASE across all warehouses. Each scenario card runs from the draft's expected service (gray) to the plan's (iris), against the dotted per-block goal.

The Overview states the answer in one sentence, then shows labor cost, hours, expected service, blocks below the goal and blocks over capacity, each against the draft. The expected service tile defines itself: the average chance a block meets its wait target. Below, a card for each scenario runs from the draft's expected service to the plan's on a full scale, with the per-block goal marked; clicking a card switches to that scenario. Further down, hours and expected service by day show where the week is tight, and alerts link each problem day to the Playbook.

Schedule: the weekly board

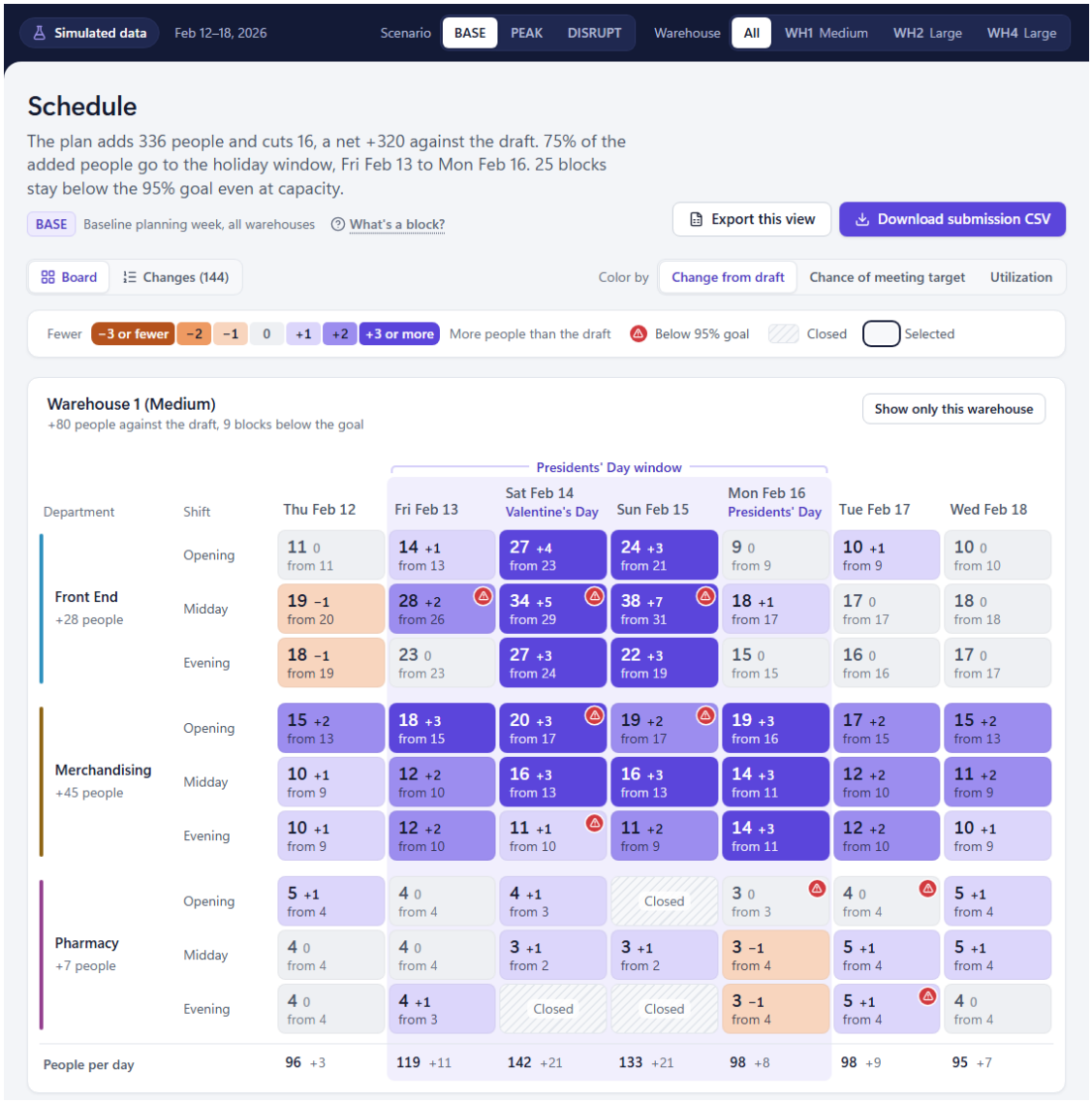


Figure 18. The Schedule page in BASE for all warehouses, showing Warehouse 1's section of the board, colored by change from the draft. The shaded columns are the Presidents' Day window.

The Schedule page is the plan itself. Its headline counts the people the plan adds and cuts against the draft, the net change, and the blocks that stay below the goal. The board has one cell per block, with departments and shifts down the side and days across the top; with all warehouses selected, each warehouse gets its own section. Each cell shows the plan's headcount, its change from the draft, and the draft it came from. The colors can show the change from the draft, the chance of meeting target, or utilization.

How to read the board. Iris cells get more people than the draft, darker for bigger changes, up to three or more; bronze cells get fewer, down to three or fewer; gray cells are unchanged. A red triangle marks a block below the service goal even at capacity. Hatched cells are closed. The row under each board totals people per day.

Block detail: how a recommendation explains itself

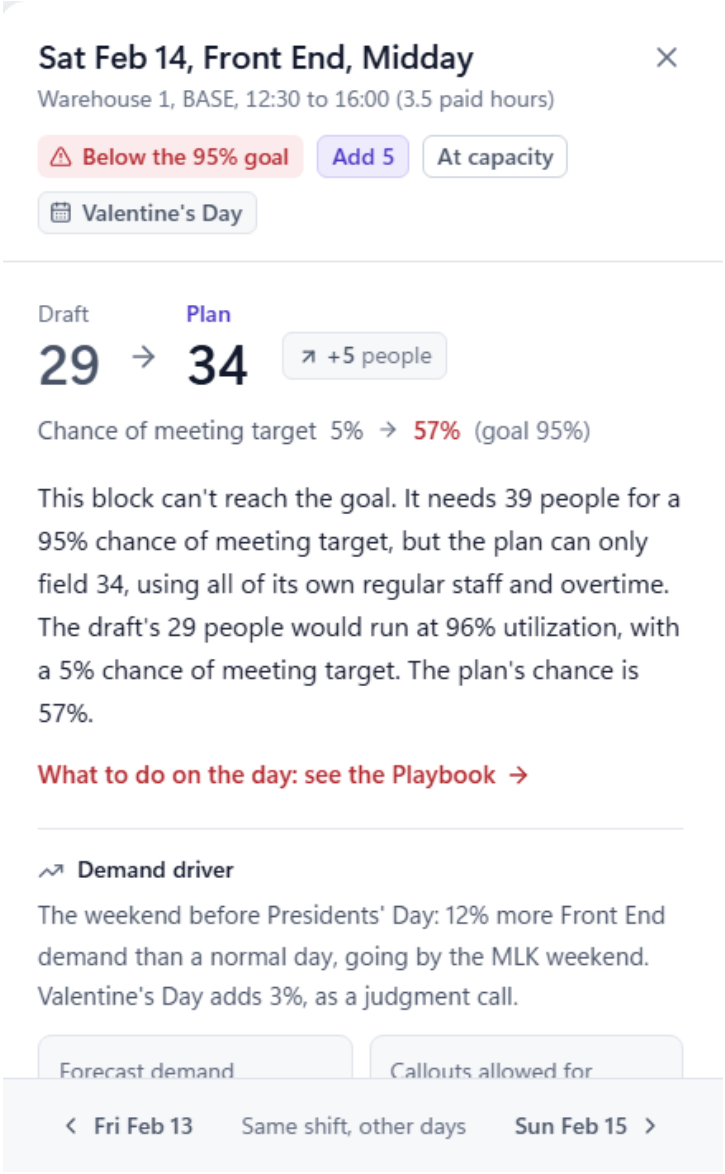


Figure 19. The detail drawer for Warehouse 1 Front End on Saturday, Feb 14, midday: a block that can't reach the goal even with all its own staff and overtime.

Clicking any cell opens its detail drawer. It gives the draft and plan headcounts, and the chance of meeting target under each, against the goal. Then comes the reason in plain words. For a block that can't reach the goal, a single link leads to what to do on the day in the Playbook. Below that are the demand driver, the forecast demand and the callout rate allowed for, and where the staff come from: regular, cross-trained or overtime, against what the block can field. Further down are its cost against the draft, its cushion, and the assumptions it rests on. The reason is written from a template filled with that block's own numbers, so it can't invent anything (D-08). Arrows at the bottom step through the same shift on other days.

Switching scenario or warehouse redraws the board. In DISRUPT at Warehouse 2, the illness forces cuts, and a badge marks every block where the draft schedules more people than can be fielded. The **Changes** tab lists every block that differs from the draft, with filters for added or cut blocks and for each department.

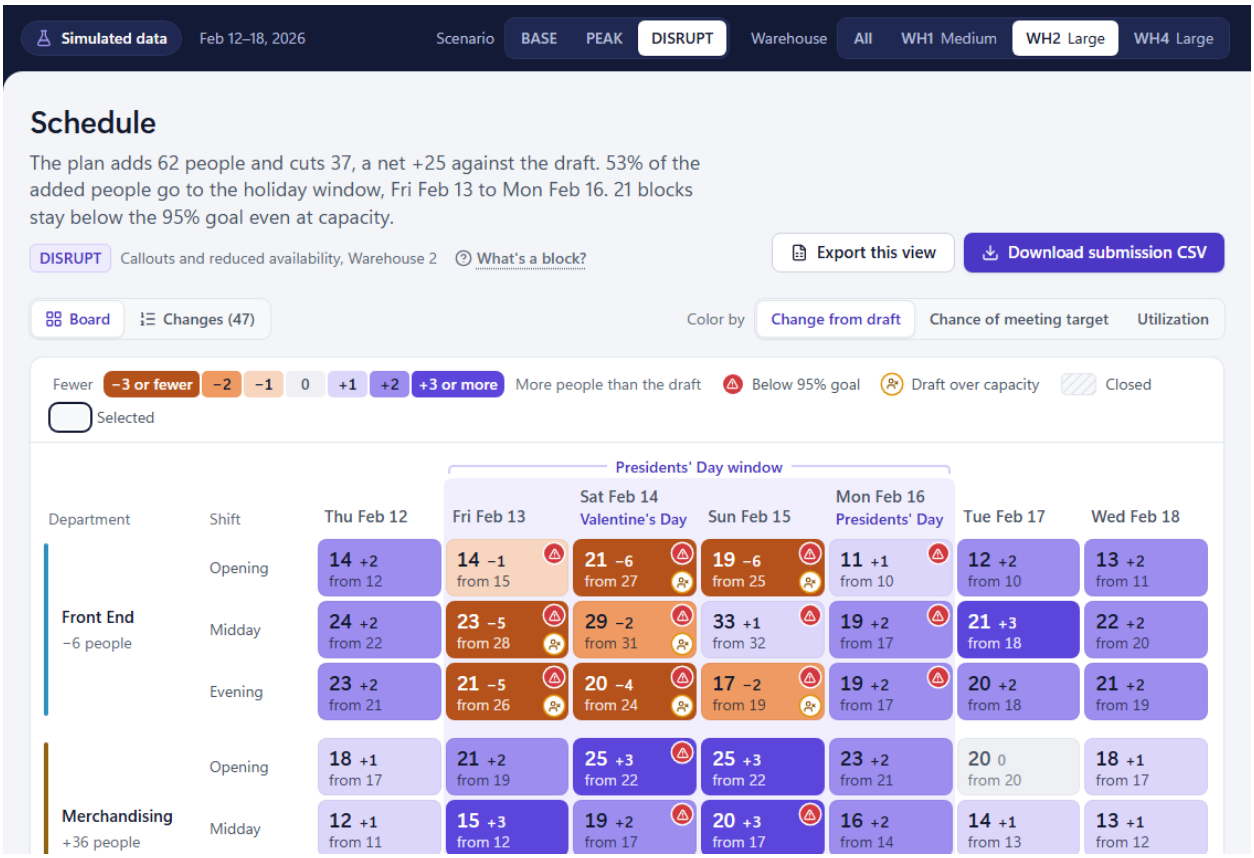


Figure 20. DISRUPT at Warehouse 2: the illness forces cuts, and a badge marks each block where the draft schedules more people than can be fielded.

Demand: when it rises and falls

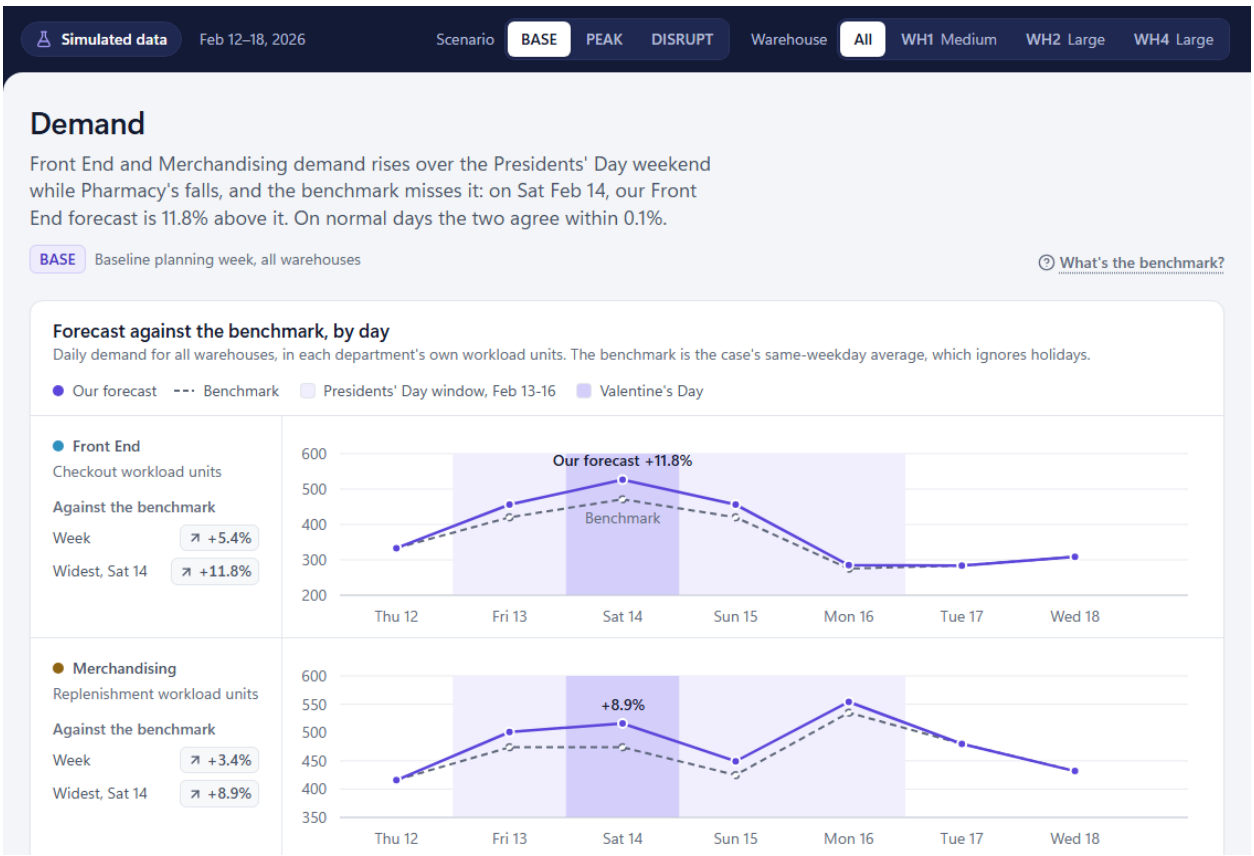


Figure 21. The Demand page in BASE: our forecast against the benchmark, by day, for each department.

The Demand page shows our forecast against the case's benchmark for each department, with the holiday window and Valentine's Day shaded, and how far apart the two are. Further down, it breaks one day's demand into its parts (a normal day, the holiday effect, Valentine's Day, and the scenario's surge), shows the shift split, and lists the drivers we tested, as in chapter 4.

Cost and service: what it costs and what it buys

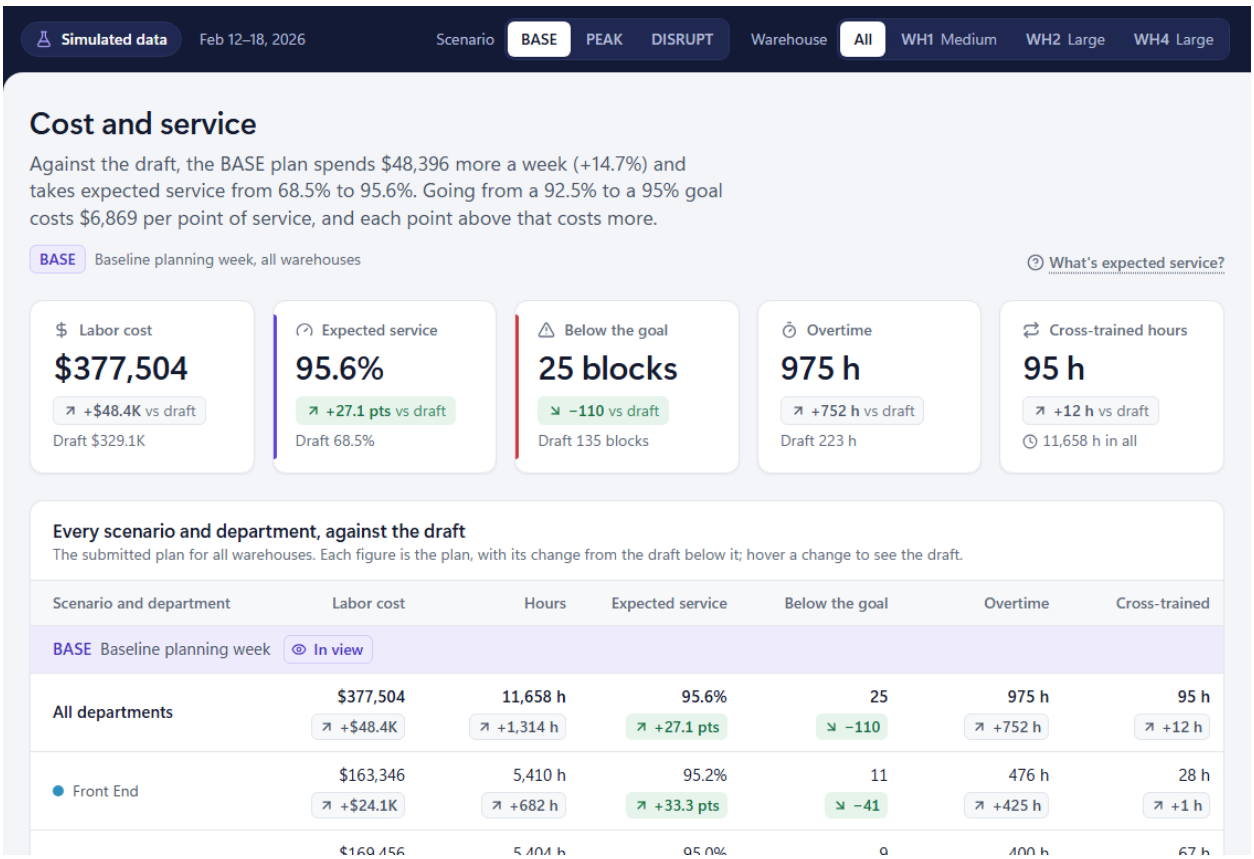


Figure 22. The Cost and service page, opening with BASE's totals against the draft.

This page sets every scenario and department against the draft in one table: labor cost, hours, expected service, blocks below the goal, overtime and cross-trained hours. Cost changes are shown in neutral gray, because spending more is neither good nor bad on its own. Below the table, the blocks below the goal are listed ten at a time, with a link to show them all, followed by the trade-offs from chapter 5: the service goal, and what the callout allowance costs.

Scenario lab: running a what-if

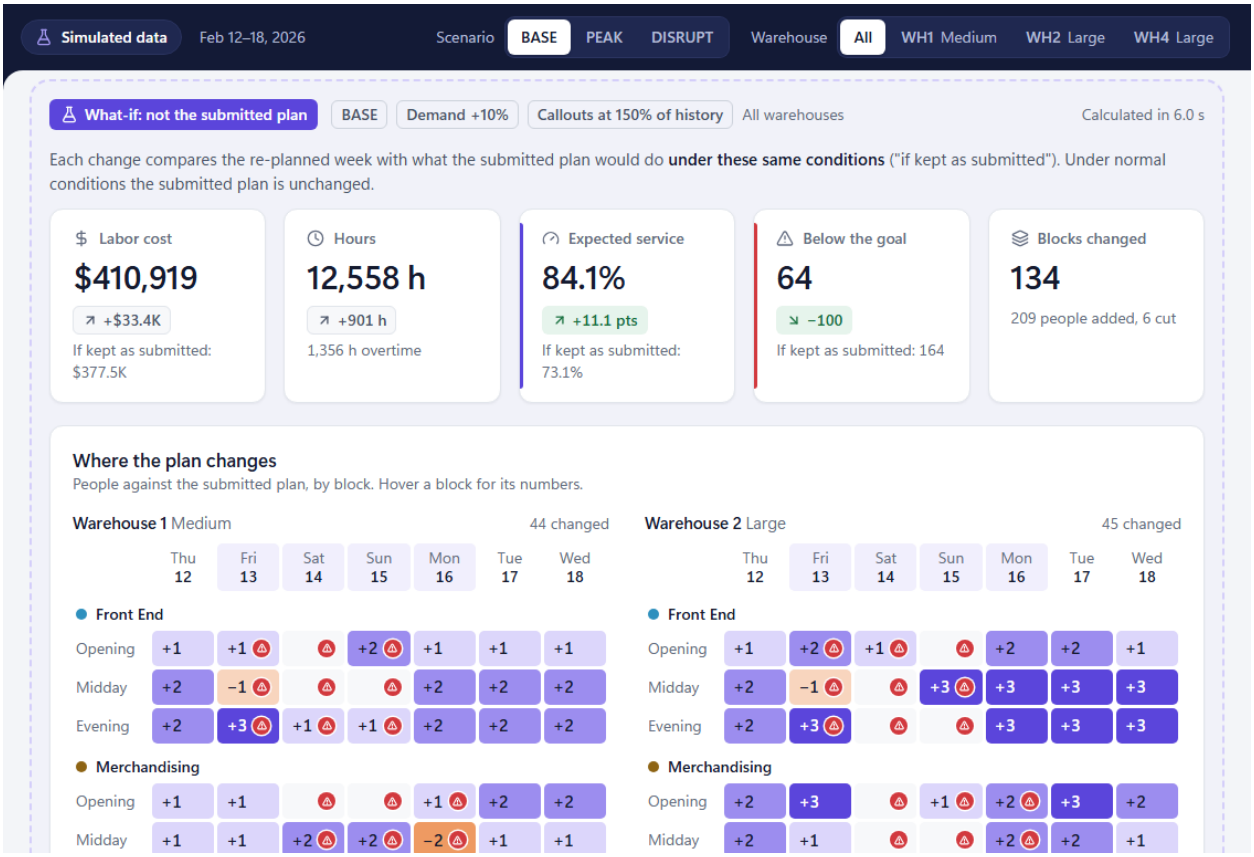


Figure 23. A what-if on BASE with more demand and more callouts than history. Each tile compares the re-planned week with the submitted plan kept as it is under the same conditions.

The Scenario lab re-plans the week live under new conditions: a different service goal, callouts as a multiple of history, demand above or below the forecast, or fewer regular staff. It reruns the forecast, the search and the risk model for every block, which takes a few seconds. Each tile then compares the re-planned week with the submitted plan kept as it is under the same conditions, labeled "If kept as submitted", which answers "what if we don't re-plan?". A map shows every block that changes. The submitted plan itself never changes, and **Reset** brings it back. Quick-start buttons set common cases, such as a busier week or an illness wave.

Playbook: what to do

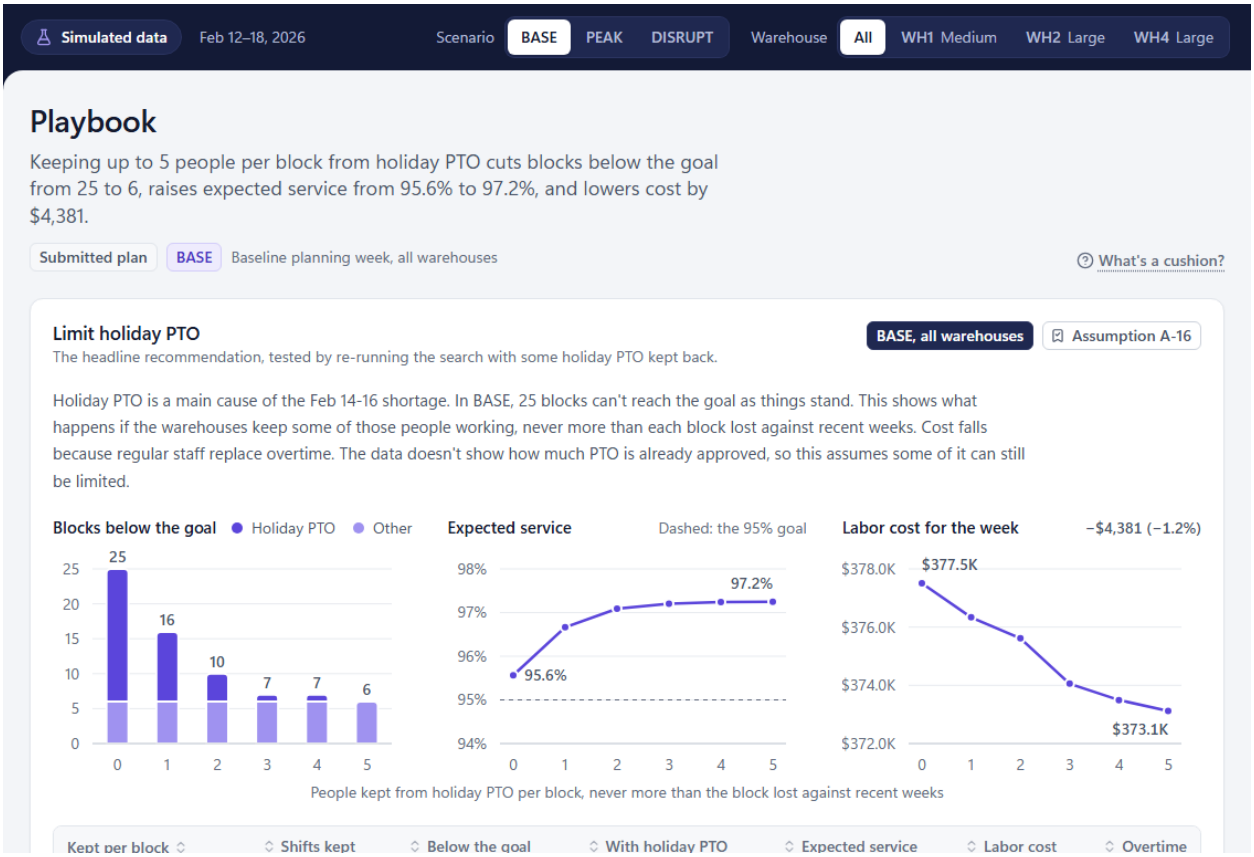


Figure 24. The Playbook page, opening with the holiday PTO test, labeled BASE across all warehouses.

The Playbook holds chapter 7's actions and follows the scenario and warehouse chosen in the top bar. It opens with the holiday PTO test, which is always run on BASE across all warehouses and says so, then shows which plan to run and when, and the rules for the day. Links jump to the Schedule with the right scenario and day selected.

Methods: how it works

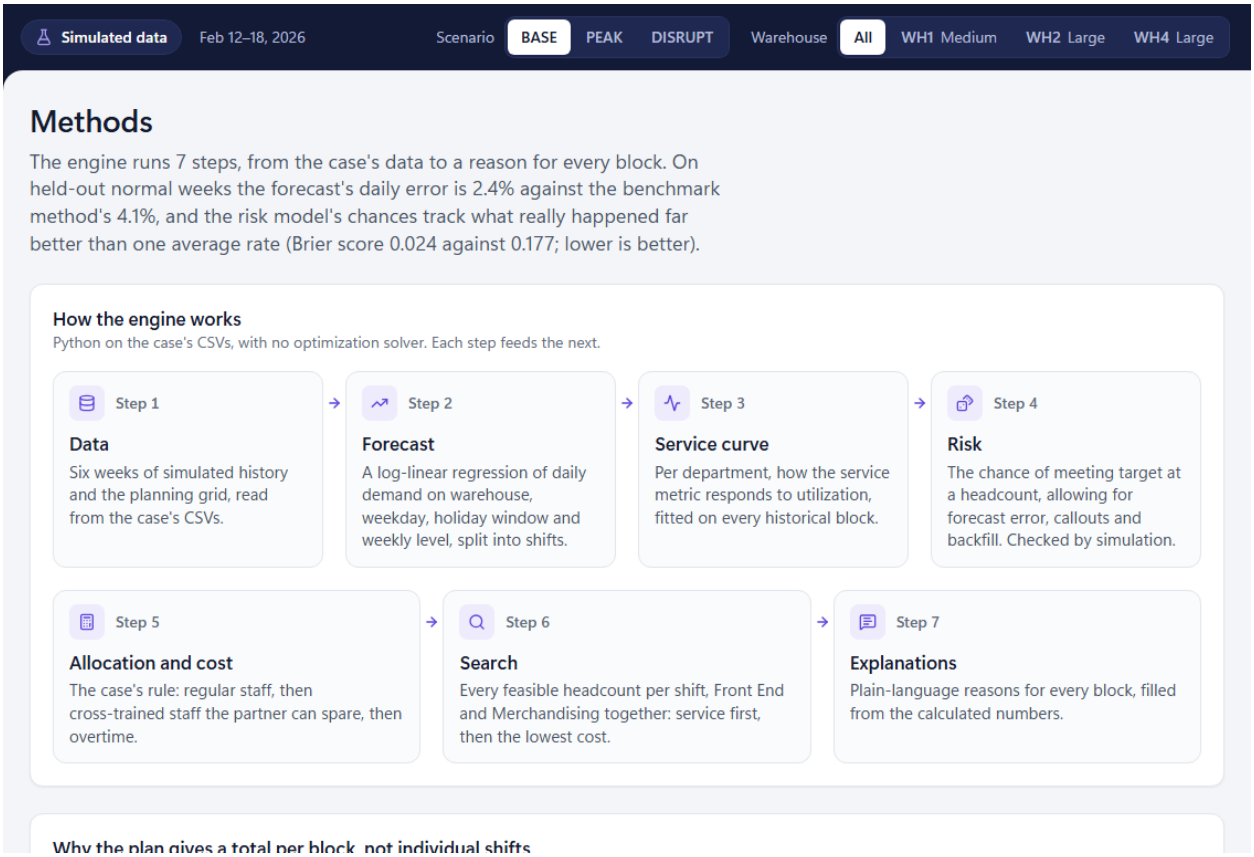


Figure 25. The Methods page: the engine's seven steps, from the case's data to a reason for every block.

The Methods page explains the engine in seven steps, from the case's data to a reason for every block, and why the plan gives totals per block. It then shows how well the model holds up on data it didn't see, and lists every assumption, with ours marked "Our judgment" apart from the case's own.

Exporting the plan

The **Download CSV** button in the sidebar saves the completed submission template with every block in every scenario, exactly as submitted. On the Schedule page, **Export this view** saves the board's blocks for the selected scenario and warehouse, with the plan, the draft, staff sources, cost and chance of meeting target for each.

9 How we checked it

A plan is only as good as the work behind it. We checked the analysis in six ways, and we list what we couldn't check.

Automated tests

The code has 121 automated test cases in 10 files. They cover the cost rule, the forecast, the service curves and risk model, the search, the submission check, the explanations, the playbook, and the app and its server. All of them pass on the frozen numbers.

Independent recomputation

Separate scripts rebuilt the key results without importing the code they check:

- **The analysis reproduces exactly.** Rerunning the pipeline rewrites every output file byte for byte (F-66).
- **The submission passes an independent rebuild:** all 567 rows, joined on keys, whole numbers, closed blocks at zero, and every open block at or above minimum staff and within capacity. 154 open blocks sit exactly at capacity (F-67).
- **The search finds the best staffing under its rule.** A brute force over every feasible pair agreed with it on every shift (F-70).
- **The cost rule matches the case's README to the half cent** on every block and every total (F-71).
- **Every explanation quotes its own block's numbers.** A test checks the draft, plan, change, utilization, chance, cost and cushion in every row's text (F-72).
- **A separate recomputation from the raw files** matched every table it could rebuild. Its service model shared our structure, so its agreement supports the arithmetic, not the assumptions (F-54).

Backtests

The forecast was tested on held-out weeks (chapter 4). The service curves were tested the same way, fitting on five weeks and predicting the sixth, six times over. Their Brier score, a standard measure of how good predicted chances are (lower is better), was 0.024, against 0.177 for simply using the average rate.

Calibration: do predicted chances come true?

A chance is only useful if blocks predicted at, say, 90% meet target about that often. Figure 26 groups history's blocks by their predicted chance, each predicted by a curve fitted without that block's week, and compares with what happened.

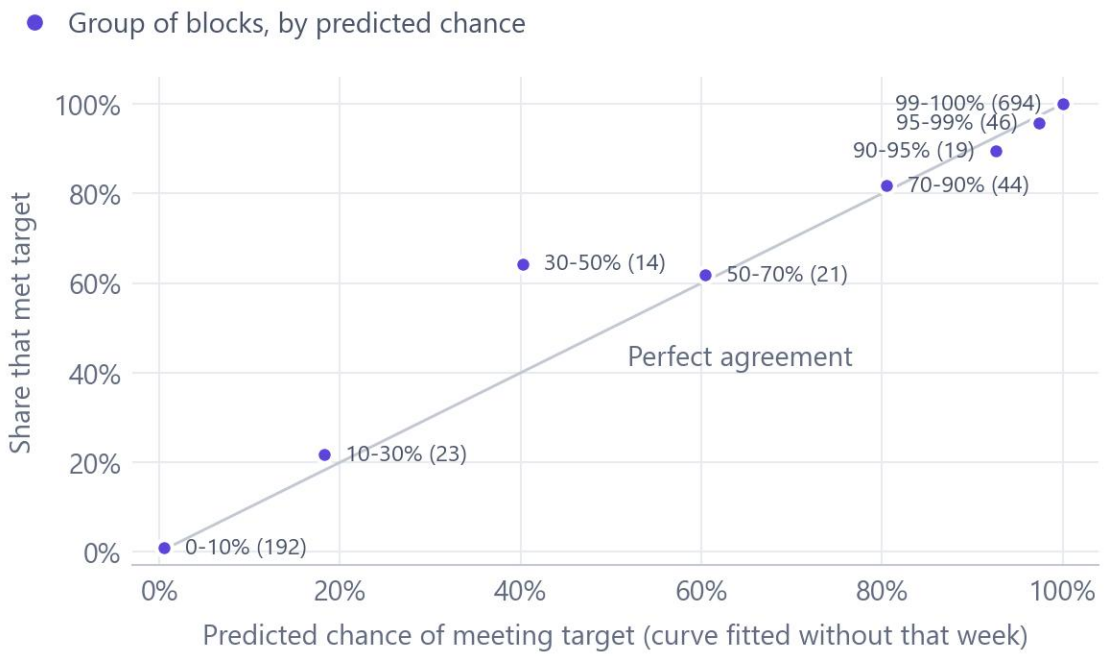


Figure 26. Predicted chance of meeting target against the share that met it, out of sample. Each dot is a group of blocks, labeled with its range and the number of blocks. Dots on the line mean predictions came true.

Predictions track outcomes. Blocks predicted at 95-99% met target 95.7% of the time, against 97.4% predicted. Blocks at 70-90% met it 81.8% of the time, against 80.5%. The first curve ran slightly optimistic near the goal; the smooth bend fixed that (D-29). As a final check, we rescored the plan using history directly instead of the curve, and expected service agreed to within half a point (F-65).

Sensitivity

We changed each judgment that could move the plan and measured the effect:

- **Valentine's Day uplift:** planning for more barely helps, because capacity binds that day (chapter 4).
- **The service goal:** each extra point costs more than the last (chapter 5).
- **The callout allowance:** worth it only if callouts happen, which history says they do (chapter 5).
- **Holiday PTO:** the strongest lever we found (chapter 7).
- **The illness callout rates in DISRUPT (A-12):** dropping them changes no staffing, because those blocks are already at capacity. Only DISRUPT's reported service moves (F-69).
- **Pharmacy's thin day-after-holiday rate (A-15):** it drives most of the Feb 17 shortfall (chapter 6).

Browser checks

The tool was driven in a real browser by an automated script, through every page, all three scenarios, the block drawer and a what-if. The screenshots in chapter 8 come from that run. The checks look for errors on the page and confirm that nothing is requested from outside the laptop.

What hasn't been checked

- **How judges score service.** The case gives only the order of priorities: service, then cost, then feasibility.
- **The holiday forecast itself.** The planning week's true demand is hidden, and borrowing the MLK weekend for Presidents' Day is an assumption (A-05).
- **The exact cross-trained staff rule.** The case doesn't publish it; ours is our reading of the README (A-01).
- **How much holiday PTO is still open to change.** The data doesn't show it (A-16).
- **The model against real warehouses.** All data is simulated. The method would need refitting on real data before use.

10 Assumptions and limits

The case asks every team to identify the assumptions it adds. Ours are listed here, apart from the case's own, together with the limits of the work and how it could run for real.

Assumptions we added

Table 17. Our assumptions. Proposed means reasoned but not confirmed by the organizers.

ID	Assumption	Why	If it's wrong	Status
A-01	Usable cross-trained staff is the smaller of the block's cross-trained pool and the partner department's regular staff left over in that shift	Our reading of the README's rule	Plans could cost more or be flagged infeasible	Proposed
A-02	Sharing follows the shift code, though the opening shift covers different hours by department	The README says "same shift"	Less sharing may be possible in opening shifts	Proposed
A-03	Plans allow for callouts at historical rates (5.2% overall), with about 49% replaced by overtime	History; the starter query lists a missing callout buffer as a weakness	If judges ignore callouts, the allowance adds cost without credit	Confirmed by the team
A-04	85% utilization is a ceiling, not a target; each department and warehouse gets its own limit	The service cliff	Too cautious adds cost; too tight misses targets	Proposed
A-05	The MLK window is a fair guide to the Presidents' Day window	The same holiday shape, Friday to Monday	The holiday effect is misjudged	Proposed
A-06	Valentine's Day adds 3% for Front End and Merchandising on Feb 14	No history; a stated judgment, tested for sensitivity	Saturday demand is misjudged, though capacity limits that day anyway	Proposed
A-07	The New Year rebound gets its own effect; the flu cluster affects callouts and availability, not demand	Demand fits almost exactly with no flu term	The baseline forecast is skewed	Confirmed
A-08	Planning-week temperature and promotion fields are taken as given	No longer needed: the forecast doesn't use them	No effect	Rejected
A-09	The historical service curve applies to the planning week	The case's docs say to estimate it from history	Service risk is misjudged where utilization was rarely seen	Proposed
A-10	Borrowed staff cost the regular rate of the department they cover	The case's docs and starter query use the block's own rate	Small cost differences	Proposed
A-11	Borrowing a holiday effect adds forecast uncertainty on Feb 13-16: 2% for Front End and Merchandising, 5% for Pharmacy	The gap between the only two comparable holiday windows	Holiday-week headroom is too small or too large	Proposed
A-12	DISRUPT's illness has January's flu-week callout rates (12.3% Front End, 10.7% Pharmacy) on top of the staff cuts	Both appeared together in January	Changes only DISRUPT's reported service, not its staffing	Checked
A-13	The service goal is a 95% chance of meeting target in each block	Service is judged first, and cost per point climbs fast above it	Cost and service move along the trade-off	Confirmed by the team

ID	Assumption	Why	If it's wrong	Status
A-14	A future week's demand level matches the last 4 weeks' average	Slightly better than all weeks in the backtest	All forecasts shift together	Proposed
A-15	Callout rates by situation are used as measured, even where few people stand behind them	Simple and traceable	A few Pharmacy blocks are over- or under-buffered, most on Feb 17	Proposed
A-16	Some holiday-weekend PTO can still be limited, up to what each block lost against recent weeks	The grid says "PTO requests", but not how many are approved	The PTO savings shrink or vanish	Proposed

The case's own assumptions

The case fixes these, and we use them as given (data pack docs/ASSUMPTIONS.md): labor hours per workload unit for each department, regular and overtime pay rates, shift times and paid hours, minimum staff per block, availability per scenario, the scenario multipliers, the service targets, and the benchmark's definition.

Limits

- **Thin holiday evidence.** History holds one holiday weekend. The Presidents' Day forecast borrows it, and Valentine's Day has no history at all.
- **Simulated data.** The demand, staffing and service relationships come from the case's simulation, not from real warehouses.
- **Totals, not names.** The plan gives each block's total and its split by source. Assigning individual people stays with managers.
- **One week at a time.** The search treats each shift separately, because weekly hour limits are already built into availability. A rule spanning several shifts would need a different method.
- **Unknown scoring.** If judges score average waits rather than pass or fail, blocks far over the cliff would matter more than our measure suggests.

How Costco could run this for real

The engine and its interface are built so that a weekly cycle could replace the case's files with live data:

1. **Feed the data weekly.** Load last week's actual demand, schedules, callouts and service, and next week's availability, holidays and planned events, in the same tables the case uses.
2. **Rerun.** Refit the forecast's weekly level and holiday effects as more holidays build up, refit the service curves, and rerun the search. This takes minutes.
3. **Review.** Leaders open the board, check the alerts and the blocks below the goal, decide on time-off approvals, and test known events in the Scenario lab.
4. **Publish.** Export the block totals to the scheduling system, where managers assign people.

5. **Learn.** Each week, compare the forecast with actual demand and predicted chances with outcomes, as in chapter 9, and investigate any drift.

The server already separates the engine from the interface through a documented API, so swapping the case's CSV files for a database connection would change one layer, not the whole tool.

A Output files and fields

Running the pipeline writes these files to the outputs folder. The report, the deck, the notebook and the app all read them, so they always agree.

Table 18. Files written by the pipeline and the chart script.

File	What it holds
TEAM_SUBMISSION_TEMPLATE_filled.csv	The submission: every block in every scenario, with total employees
plan_vs_draft.csv	Every block: demand, plan, draft, staff sources, capacity, cost, utilization, chance of meeting target, and whether it is below the goal
explanations.csv	Every block: the action, the reason, the demand driver, cost and service effects, the cushion and the assumptions it rests on
capacity_gaps.csv	People short of the goal by scenario, day and department
goal_tradeoff.csv	Cost and expected service at each service goal
pto_what_if.csv	The holiday PTO test
forecast_daily.csv, forecast_blocks.csv	The forecast by day and by block, with the benchmark
service_limits.csv	Utilization for a high, medium and even chance of meeting target, by department and warehouse
diagnosis_*.csv	The historical diagnosis: staffing gaps, callout rates and service by utilization
analysis_summary.json	Every headline figure, read by this report
charts/*.png, charts/figure_data.json	This report's charts, and the figures they are drawn from

The main fields of plan_vs_draft.csv: **recommended** is the plan's total employees; **draft_scheduled_employees** is the draft; **fill_regular**, **fill_flex** and **fill_overtime** split the plan by source; **capacity** is the most the block can field; **cost** follows the case's rule; **planned_utilization** and **chance_met** are the plan's utilization and chance of meeting target; **below_service_goal** flags blocks that can't reach the goal; **unfilled** counts draft staff who can't be fielded.

B The submission format and its validation

The submission template has 567 rows: every warehouse, department, day and shift in all three scenarios. We fill **submitted_employees** with each block's total employees, and 0 for the 27 closed blocks. Rows are joined to the plan on their keys (scenario, warehouse, department, date and shift), never by row order, because the template sorts shifts alphabetically.

Before saving, a validator checks that every row is present and matches its keys, that every value is a whole number, that closed blocks are zero, and that every open block is at or above minimum staff and

within capacity. On the frozen plan it found 0 problems. Its final mode also rejects the placeholder team name, which must be replaced before submitting:

```
# First set TEAM_ID in src/submission.py, then rebuild
python -m src.pipeline
# Then the final check prints Passes, or the problems it found
python -c "import pandas as pd; from src.submission import validate;
print(validate(pd.read_csv('outputs/TEAM_SUBMISSION_TEMPLATE_filled.csv',
parse_dates=['day_date']), final=True) or 'Passes')"
```

C How to reproduce everything

After the setup steps in chapter 8, these commands rebuild every result from the case's files. The pipeline takes a minute or two. The documents need Node.js and one npm install.

```
python -m pytest                    # every test
python -m src.pipeline              # the analysis: writes outputs/
python -m src.charts                # this report's charts: writes outputs/charts/
python planning/evidence/ws1_checks.py --full    # the independent checks
node scripts/build_analysis_doc.js   # this report, its claims register and its PDF
```

Every calculated figure in this report is read from those outputs by the report's builder, never typed. The builder records each figure with the file and key it came from in planning/evidence/report_claims.json, and refuses to build if any number lacks a source. Where a figure is a fixed fact of the case, such as a service target, the register cites the case document instead.

D Glossary

Term	Meaning
Block	One warehouse, department, day and shift: the unit the plan is made in.
Draft	The manager's draft schedule for the planning week, given by the case.
Scenario	One set of conditions for the week: BASE (expected), PEAK (more demand) or DISRUPT (fewer staff).
Workload units	Each department's own measure of demand. Units are never compared across departments.
Utilization	The hours of work a block's demand needs, divided by the hours its staff actually work.
Meets target	A block whose wait or response time is at or under its department's target.

Term	Meaning
Service curve	How the chance of meeting target falls as utilization rises, fitted on history for each department.
The cliff	The narrow range of utilization, near 85-90%, where service breaks down.
Chance of meeting target	Our model's estimate for one block, allowing for forecast error and callouts.
Expected service	The average chance of meeting target across open blocks.
Service goal	The chance of meeting target the plan aims for in every block.
Callout	A scheduled employee who doesn't come in. Backfill replaces some of them with overtime.
Cross-trained staff	Front End and Merchandising employees who can work in the partner department in the same shift.
Capacity	The most people a block can field: regular staff, usable cross-trained staff and overtime.
Cushion	The people a block can lose, or the extra demand it can take, before missing target becomes likelier than not.
Benchmark	The case's naive forecast: the average of the same weekday over the last four weeks of history.
Holiday window	Friday to Monday around a holiday: the MLK weekend in January, the Presidents' Day weekend in the planning week.
PTO	Paid time off. Holiday-weekend PTO requests reduce available staff on Feb 14-16.
Backtest	Testing a forecast on past weeks it wasn't fitted on.
Brier score	A measure of how good predicted chances are: the average squared gap between prediction and outcome. Lower is better.